

Master's Programme in Electronics and Electrical Engineering

IoT Device Authentication in 5G Networks

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Abstract

The rapid expansion of the Internet of Things and Industrial IoT has accelerated the adoption of distributed sensing, automation, and data-driven decision-making across critical sectors such as manufacturing, energy, healthcare, and logistics. IoT devices typically possess limited computational resources, heterogeneous hardware architectures, and diverse communication protocols, making secure and scalable integration into modern communication infrastructures a persistent challenge. Although 5G provides high reliability, ultra-low latency, and massive connectivity—factors which makes it a promising foundation for IoT systems—current integration approaches often rely on complex architectures, proprietary gateways, and heavyweight security frameworks that limit their practicality in resource-constrained and industrial environments. To address these limitations, this thesis presents a lightweight and interoperable IoT-to-5G integration framework featuring a minimal User Equipment and gNB gateway stack implemented in C. The system can operate directly on embedded IoT devices such as Raspberry Pi and non-cellular IIoT platforms like DECT-2020 NR, e.g. Nordic Semiconductor nRF91 family. Moreover, the UE and gNB gateways are designed to connect seamlessly with the private 5G Core Networks for device authentication, enabling secure, scalable, and standards-compliant integration across heterogeneous IoT environments. Experimental validation demonstrates that the designed UE and gNB gateway interoperate seamlessly with both commercial and open-source 5G systems, including CumuCore, OpenAirInterface, and free5GC. Unlike traditional IoT gateways that locally handle protocol translation and security overhead, the proposed design offloads authentication, session management, and traffic control directly to the 5GC, thereby reducing the computational burden on edge IoT and IIoT devices while ensuring full compliance with 3GPP standards.

Keywords IoT , IIoT , Authentication, 5G Core, Private 5G, 3GPP

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List of Abbreviations

AAA	Authentication, Authorization, and Accounting
AMF	Access and Mobility Management Function
ARPF	Authentication credential Repository and Processing Function
AUSF	Authentication Server Function
AUTN	Authentication Token
B5G	Beyond 5G
CK	Cipher Key
CTR	Counter mode
D2D	Device-to-Device
DoS	Denial of Service
eMBB	Enhanced Mobile Broadband
EAP-AKA	Extensible Authentication Protocol – Authentication and Key Agreement
EAP-AKA'	Extensible Authentication Protocol – Authentication and Key Agreement Prime
gNB	Next Generation Node B
HPLMN	Home Public Land Mobile Network
HSS	Home Subscriber Server
IE	Information Element
IIoT	Industrial Internet of Things
IK	Integrity Key
IoT	Internet of Things
IDS	Intrusion Detection System
IP	Internet Protocol
KDF	Key Derivation Function
MAC	Message Authentication Code
MCC	Mobile Country Code
mMTC	Massive Machine-Type Communications
MNC	Mobile Network Code
N3IWF	Non-3GPP Interworking Function
NEF	Network Exposure Function
NR	New Radio
NRF	Network Repository Function
OAI	OpenAirInterface
PCF	Policy Control Function
PER	Packet Error Rate
PLMN	Public Land Mobile Network
SA	Security Association
SBA	Service-Based Architecture
SCTP	Stream Control Transmission Protocol
SMF	Session Management Function
SNPN	Standalone Non-Public Network
SoftSIM	Software-based SIM
SUCI	Subscription Concealed Identifier

SUPI	Subscription Permanent Identifier
5G	Fifth Generation (mobile network)
5G-AKA	5G Authentication and Key Agreement
5GC	5G Core
TNGF	Trusted Non-3GPP Gateway Function
TSN	Time-Sensitive Networking
UDM	Unified Data Management
UE	User Equipment
UPF	User Plane Function
URLLC	Ultra-Reliable Low-Latency Communications
WAT	Wireless Authentication Technology
RAND	Random Challenge
URL	Uniform Resource Locator

1 Introduction

The rapid growth of the Internet of Things (IoT) and its industrial counterpart, the Industrial Internet of Things (IIoT), is transforming domains, such as automation, manufacturing, healthcare, logistics, as well as critical infrastructures. As illustrated in Fig. 1, the number of connected IoT devices worldwide has grown dramatically, rising from 12.6 billion in 2015 to a projected 33.4 billion in 2025—nearly tripling within a decade. By enabling interconnected devices to exchange, process, and act on data, IoT technologies contribute to increased efficiency, productivity, together with intelligent decision-making in many fields [1–3]. In the industry, IIoT has become a key enabler of the Fourth Industrial Revolution (Industry 4.0) through the integration of distributed sensing, monitoring, and control [4, 5]. This integration helps enhance productivity as well as mitigates the need for human intervention, thereby increasing revenue while reducing costs.

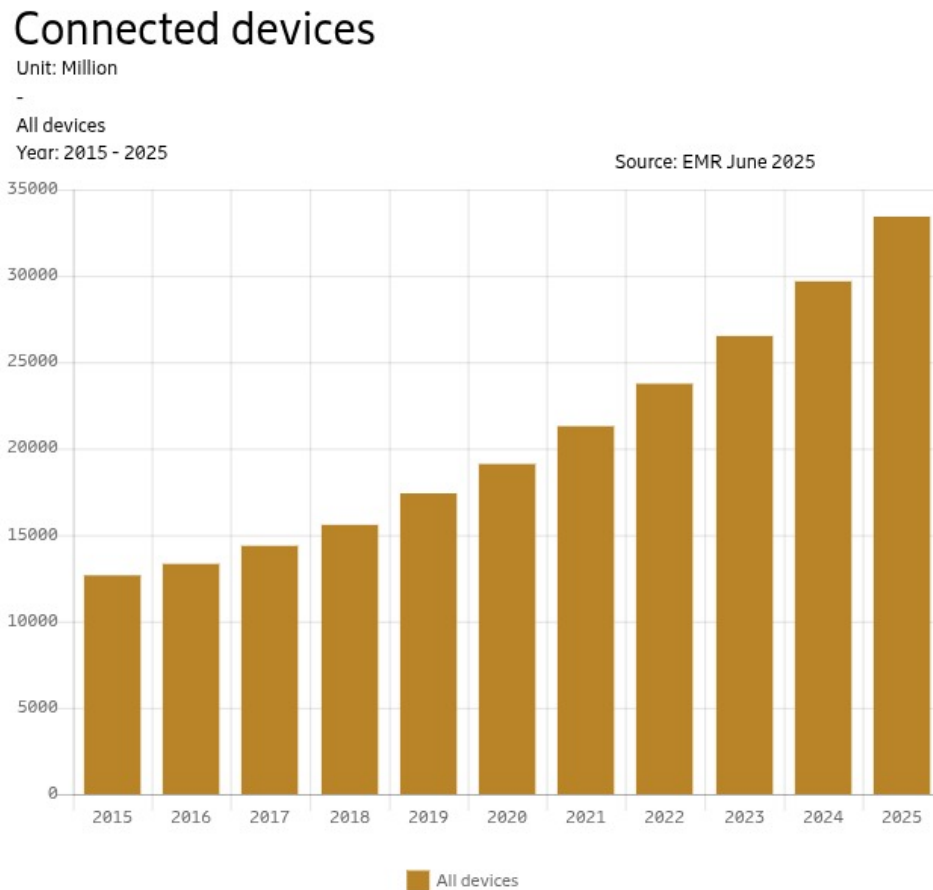


Figure 1: Report of total number of connected devices worldwide from 2015-2025 from Ericsson. Adapted from [6]

Despite these benefits, IoT/IIoT devices remain highly resource-constrained, heterogeneous, as well as exposed to evolving cyber-physical threats. Security, privacy, and trust are critical for IIoT where the entire industrial ecosystem can be damaged

due to a single device malfunction. To address these risks, various mechanisms have been proposed. Intrusion Detection Systems (IDS) have been widely adopted to monitor, analyze, and detect malicious behavior in IoT/IIoT networks [4]. Recent trends focus in machine learning, deep learning, together with ensemble methods to improve accuracy and adaptability [7–9]. Moreover, blockchain technology has been introduced to IIoT architectures as a distributed, tamper-resistant technology that enhances privacy, scalability, and security, especially when combined with 5G and edge computing [10–13]. Aside from these generic IoT approaches, researchers have increasingly focused on securing 5G-enabled IoT systems. There has been an extensive review of 5G IoT, categorizing solutions as blockchain-based solutions, cryptographic and signature-based methods, as well as graph-theoretical models [14].

While effective, these approaches often show limitations in practice due to their incompatibility with existing infrastructure and resource-constrained devices. In many IIoT deployments, wide-area connectivity is provided by cellular networks, while local connectivity relies on short-range technologies such as Wi-Fi or DECT-2020 NR [15]. In these non-public wireless networks, constrained IoT devices are often responsible for critical functions including authentication, end-to-end security, or session management using local credentials and ad-hoc trust schemes [16, 17].

There is a need for architectures that allow IoT devices using non-cellular access technologies to leverage standardized 5G security mechanisms, improving both scalability and security in IIoT deployments. To address this gap, this thesis first reviews how an IoT device can register with a private 5G Core (5GC) via non-3GPP access. Correspondingly, we then aim to propose and implements a minimal IoT User Equipment (UE) and gNB gateway stack in C, tailored for IoT/IIoT devices. The proposed system need to enable IoT devices to register directly with commercial and open-source 5G Cores, offloading authentication, session management, and traffic control to the 5GC rather than performing these tasks on the device or gateway. Finally, the thesis experimentally evaluates whether the implemented design can successfully interoperate with standardized 5G procedures while reducing device-side complexity, demonstrating a scalable and practical integration model for local-area deployments such as industrial automation and private 5G networks.

This thesis is structured as follows: Section 2 presents the background and fundamentals, exploring the architecture and security challenges of IoT/IIoT, as well as the authentication protocols of 5G core network. Section 2 also reviews literature related to the thesis topic including intrusion detection systems, blockchain-enabled IIoT/IoT cybersecurity, as well as how IoT is secured in 5G. Section 3 illustrates the methodology, including the system design, choice of implementation, security architecture, together with integration with commercial or open-source 5G Core networks. Section 4 describes the implementation of the IoT UE and gNB gateway, the deployment of 5G cores, and the hardware testbed used for evaluation. Section 4.3 presents the evaluation results under various scenarios, examining system performance, registration, and authentication behavior. Finally, Section 5 concludes the thesis, discusses the findings, while considering directions for future research, including potential extensions toward 6G multi-access connectivity.

2 Literature Review

This section introduces the key concepts and technologies that form the foundation of this thesis. It begins with an overview of the IoT and its communication models, followed by an examination of resource constraints, interoperability challenges, and security threats inherent in IoT systems. Subsequently, the fundamentals of fifth-generation (5G) mobile networks and their integration with IoT are presented, emphasizing the architectural principles and their significance for the IIoT. The discussion then turns to the security frameworks within 5G-enabled IoT, focusing on the 5G Authentication and Key Agreement (5G-AKA) and the Extensible Authentication Protocol-based AKA' (EAP-AKA') mechanisms, as well as the security considerations for non-3GPP access and private 5G core integration. Furthermore, advanced IoT security solutions and state-of-the-art developments beyond 5G, such as intrusion detection and blockchain-enabled IIoT, are explored. The section concludes with a summary and discussion of the reviewed literature.

2.1 Internet of Things

The Internet of Things (IoT) continues to expand rapidly, and the 6G vision is to support massive-scale deployments by enabling multi-connectivity across heterogeneous networks while meeting critical requirements for ultra-reliable low-latency communications, high data rates, intelligent orchestration, and end-to-end security [18]. This trend has accelerated the digital transformation of Industry 4.0, embedding real-time sensing, control, and analytics into manufacturing systems, utilities, and supply chains. This industrial shift has given rise to the IIoT, a specialized IoT paradigm focused on mission-critical applications that demand high reliability, deterministic latency, and secure large-scale integration of cyber-physical systems.

2.1.1 IoT architecture and communication models

The ITU-T reference model for the IoT, defined in Recommendation Y.4000/Y.2060 [19], outlines four key layers: the device layer, network layer, service and application support layer, and application layer, as shown in Fig. 2. The device layer comprises sensors, actuators, and gateways. Devices may connect directly to the network or indirectly through gateways, which provide multi-interface support (e.g., ZigBee, Bluetooth, Wi-Fi, LTE) and protocol conversion across heterogeneous technologies. The network layer provides connectivity and transport functions, including access control, mobility management, and authentication, authorization, and accounting (AAA). It ensures reliable delivery of IoT data and control messages between devices and higher layers. The service and application support layer offers middleware functions such as data storage, aggregation, and device management, enabling interoperability between the network and application layers. Finally, the application layer hosts domain-specific IoT services—such as smart homes automation, industrial monitoring, and healthcare services—translating data into actionable information for end users. Together, these layers enable flexible communication models such as device-to-device

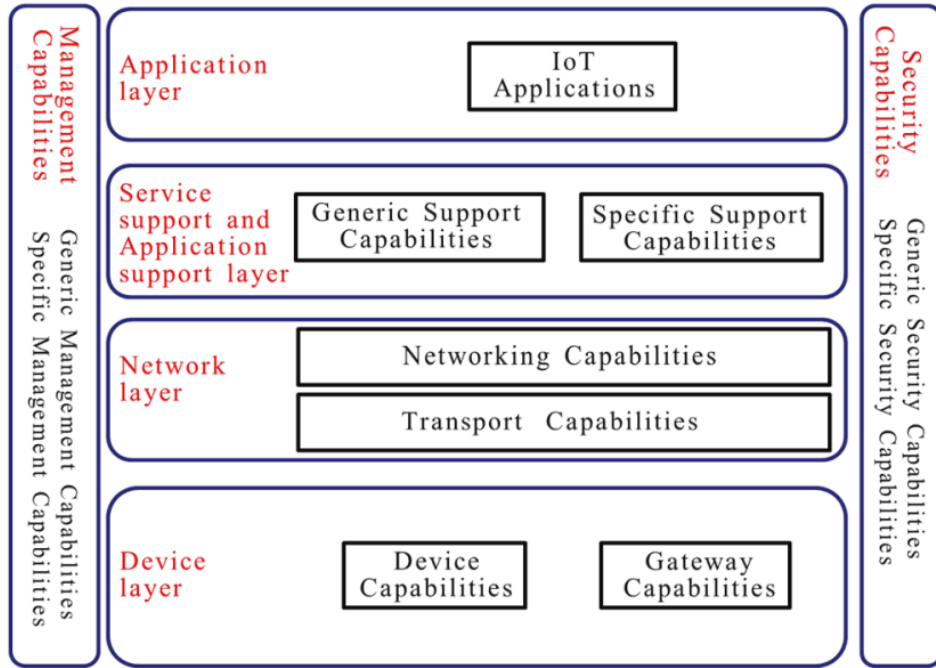


Figure 2: IoT reference model from ITU-T. Adapted from [19]

(D2D), device-to-gateway, and device-to-cloud, forming the foundation of modern IoT systems [19].

The model also specifies cross-layer management and security capability blocks. Management functions include remote device management, topology, and congestion management, while security management, such as authentication/authorization and confidentiality/integrity, provides generic protections across device/network/application layers. Additional application-specific capabilities are also considered when required [19].

Several key parameters influence the design of IoT architectures, including scalability, energy efficiency, network addressing, and security. Research has focused on [20]:

- **Scalability:** Multi-hop routing protocols and hierarchical architectures enable IoT networks to cover larger areas and adapt dynamically to network growth.
- **Energy efficiency:** Techniques such as energy harvesting, optimized MAC protocols, and cross-layer designs minimize power consumption in sensor nodes and network devices.
- **Addressing and interoperability:** The integration of IPv6 with low-power networks (6LoWPAN) and the adoption of standardized protocols allow large-scale, heterogeneous IoT deployments.
- **Security and privacy:** Cryptographic mechanisms, authentication protocols, and secure remote reprogramming (e.g., Deluge-based protocols) protect data integrity and prevent unauthorized access.

Consequently, hierarchical IoT architectures, such as the three-layer model comprising the Sensing and Control Layer (SCL), Information Processing Layer (IPL), and Application Layer (AL), have been proposed to optimize energy usage and hardware resource utilization [20]. In these architectures, sensor nodes (SNs) are managed by gateway nodes (eGNs), which coordinate with base stations (eNodes) for communication and resource allocation. Data is processed and analyzed in the cloud or edge resources, with results translated into human-readable formats and provided to end users through visualization tools. Alternative architectures, including cloud-based models, 3G-PLC integration, and self-configuring peer-to-peer networks, have also been explored to address scalability, energy efficiency, and large-scale deployment challenges. These designs collectively form the foundation for deploying Industrial IoT systems in smart manufacturing, energy, and utility networks [20].

The IIoT is a specialized subset of IoT that integrates intelligent devices and solutions into industrial processes. By embedding real-time sensing, monitoring, and control mechanisms into manufacturing, energy, and logistics, IIoT enhances operational efficiency, reliability, and scalability. The IIoT envisions a fully digitized factory floor in which sensors, actuators and control units continuously exchange data to enable real-time monitoring, predictive maintenance and adaptive manufacturing [21, 22].

In many industrial sectors—such as utilities, mining, and factory automation—real-time automation has traditionally relied on wired communication networks, most commonly based on real-time extensions of IEEE 802.3 Ethernet [23]. While these extensions enable deterministic performance, they are often incompatible with standard Ethernet, creating significant challenges for integration with enterprise IT networks. Such convergence, however, is critical for realizing the Industry 4.0 vision of fully integrated cyber-physical systems [21]. To address this gap, the IEEE 802.1 Time-Sensitive Networking (TSN) task group’s standards have emerged as an open framework that supports the coexistence of deterministic traffic with best-effort traffic over Ethernet [24]. TSN thus represents a significant step toward truly converged industrial and enterprise networks.

In parallel, several wireless technologies have been developed specifically for industrial communication [21]:

- **ISA100.11a and WirelessHART:** Both are IEEE 802.15.4-based standards designed for short-range mesh connectivity in process automation.
- **IO-Link Wireless:** An extension of the IO-Link IEC 61131-9 standard, this Bluetooth-based solution enables wireless networking between sensors, actuators, and controllers in factory automation systems.
- **Industrial Wireless LAN:** Offered by vendors such as Siemens, these solutions extend WLAN technology to industrial domains such as automotive manufacturing, transportation, and logistics.

Together, these wired and wireless solutions highlight the evolution of industrial networking from proprietary, domain-specific protocols toward standardized,

interoperable platforms capable of supporting the stringent requirements of IIoT applications.

2.1.2 Resource constraints and interoperability challenges

IoT systems are heterogeneous devices with limited resources, posing significant challenges in communication, computation, and security. Most IoT nodes, such as sensors and actuators, are constrained in energy, memory, and processing power, making the direct adoption of conventional Internet security and communication protocols infeasible [25, 26]. Another constraint lies in energy management. Many IoT devices rely on small, non-rechargeable batteries, making power efficiency critical for long-term deployments. Low-power communication protocols exist, yet frequent battery replacement remains a barrier for large-scale IoT systems. Similarly, memory limitations restrict the ability of nodes to store data or perform complex cryptographic operations, limiting system scalability and resilience [26].

Interoperability is another major challenge due to the diversity of network technologies, data formats, and communication protocols. Devices using different standards—such as ZigBee, Bluetooth, Wi-Fi, or LTE—often require gateways for protocol translation, introducing complexity and latency. The absence of a unified IoT architecture further complicates cross-platform communication and service discovery [25, 27].

2.1.3 Security threats and vulnerabilities in IoT systems

Due to the dynamic and resource-limited nature of IoT systems, they are highly susceptible to various security threats [22, 28, 29]. Resource constraints hinder the deployment of traditional encryption and authentication methods, such as key encryption and spread-spectrum techniques [22]. Common attacks—such as denial-of-service (DoS), spoofing, jamming, man-in-the-middle, malware injection, and buffer overflow—target different layers of the IoT architecture. Fig. 3 summarizes typical attacks across these layers.

At the physical layer, threats include device tampering and DoS attacks [22, 30]. Devices deployed in exposed environments are susceptible to physical modification or firmware replacement, which can subsequently be used to compromise other nodes. Spread-spectrum countermeasures are effective but often infeasible due to energy and processing limitations.

In the network layer, vulnerabilities stem from the heterogeneity and limited trust mechanisms among devices due to the lack of resources [22, 31, 32]. Weak nodes can be exploited to gain unauthorized trust, enabling indirect attacks such as man-in-the-middle (MITM) intrusions, which compromise confidentiality and data integrity through eavesdropping or message replay.

At the application layer, security risks are similar to those found in conventional software systems. Poor input validation, insecure APIs, and unpatched middleware can allow attackers to inject malicious code, steal sensitive data, or bypass authentication mechanisms [22, 33].

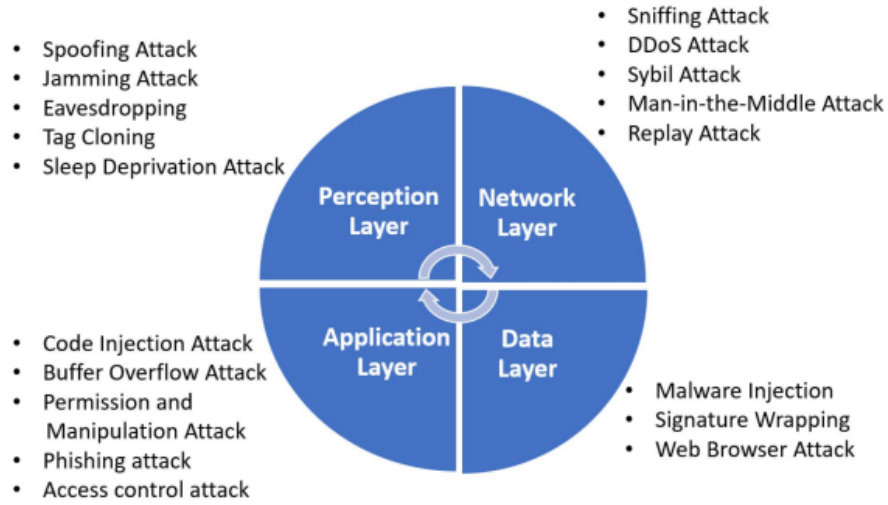


Figure 3: Attacks on layers of IoT architecture. Adapted from [29]

Traditional security models based on the CIA triad (confidentiality, integrity, availability) are often inadequate for IoT systems. Emerging concerns such as trustworthiness and privacy have become equally critical, particularly in industrial IoT (IIoT) applications where system compromise can lead to physical and operational hazards [22].

Overall, IoT security requires multi-layered defenses that consider both the unique limitations of resource-constrained devices and the complex interdependencies between layers.

2.2 5G-Enabled IoT Integration

Although projects such as 5G-CLARITY [34] showcase the potential of 5G-Advanced private networks by integrating 5G NR, Wi-Fi, and (optionally) LiFi with enhanced multiconnectivity, high-accuracy localization, and cross-wireless-access-technologies (WATs) slicing through 3GPP non-3GPP interworking functions (e.g., N3IWF for untrusted access and TNGF for trusted access), practical deployment remains limited since most public 5G cores do not yet expose or support these capabilities. Nevertheless, while multi-connectivity is essential, its realization depends on robust and unified authentication, which is currently absent as different WATs rely on distinct authentication protocols, encryption methods, and security infrastructures, thereby constraining device management and inter-WAT handovers. Furthermore, the heterogeneity of frequency bands and radio characteristics across WATs exacerbates the difficulty of designing cohesive frameworks for authentication and mobility.

In many deployments, while cellular standards are used for wide-area access, local connectivity relies on short-range technologies such as Wi-Fi or Bluetooth [16]. In these wireless local-area solutions, however, constrained IoT devices often remain responsible for key functions such as device authentication, end-to-end security,

session management, and traffic control [17]. These responsibilities can increase overhead and complicate scaling when large numbers of nodes must be managed, therefore, reducing the scalability and connectivity of the IIoT deployment. Unlike public cellular networks, which benefit from the globally standardized and secure 5G Authentication and Key Agreement (5G-AKA) protocol, non-public networks such as Wi-Fi and DECT-2020 NR typically depend on local credentials and ad-hoc trust schemes. For example, Wi-Fi authentication often depends on SSID-passphrase credentials, which are vulnerable to social engineering and rogue access point attacks, whereas DECT-2020 NR devices, despite supporting hop-by-hop AES-128 encryption and integrity protection at the MAC layer, currently lack a globally standardized authentication framework, and private deployments may rely on ad-hoc trust schemes that are less mature than 3GPP's well-established 5G-AKA system.

The fifth-generation core (5GC) natively supports secure device registration, authentication, mobility management, and traffic control [35]. By shifting these tasks from devices to the Core, resource-constrained IoT platforms can benefit from carrier-grade security and efficient traffic handling. Next section explore the fundamentals of 5GC.

2.2.1 5G architecture overview and key feature

In addition to traditional IoT architectures, alternative architectural models have been proposed to support large-scale integration with emerging wireless technologies such as 5G. One such design is the five-layer IoT architecture within a 5G framework [36], which enhances scalability, flexibility, and end-to-end connectivity from sensing devices to cloud services. As illustrated in Fig. 4, this model introduces dedicated layers for communication and network management, enabling seamless interaction between heterogeneous IoT devices through low-power wide-area networks (LPWANs) such as Sigfox, LoRa, ZigBee, and NB-IoT. Data is aggregated at IoT gateways and forwarded over 5G New Radio (NR) links to the core network for processing and application delivery across domains such as smart industry, transportation, and energy systems.

This layered approach complements hierarchical IoT models by providing native support for high-throughput, low-latency communication and massive device connectivity—characteristics essential for future IIoT deployment over 5G networks. 5G-and-beyond networks are positioned to be the unifying fabric for this vision: the 5G Core incorporates Time-Sensitive Networking to guarantee deterministic traffic, and Multi-Access Edge Computing to bring low-latency processing to the plant floor [21]. The architecture supports all three 5G service categories—enhanced mobile broadband (eMBB), massive machine-type communications (mMTC) and ultra-reliable low-latency communications (URLLC)—thus allowing a single radio interface to satisfy the diverse, heterogeneous demands of industrial applications, from high-throughput video analytics and AR/VR to mission-critical robotic control [21]. Moreover, 5G's flexible deployment options (private, public and hybrid) and spectrum sharing (licensed and unlicensed) provide plant owners with the ability to design dedicated, end-to-end secure networks that can be isolated from carrier infrastructure

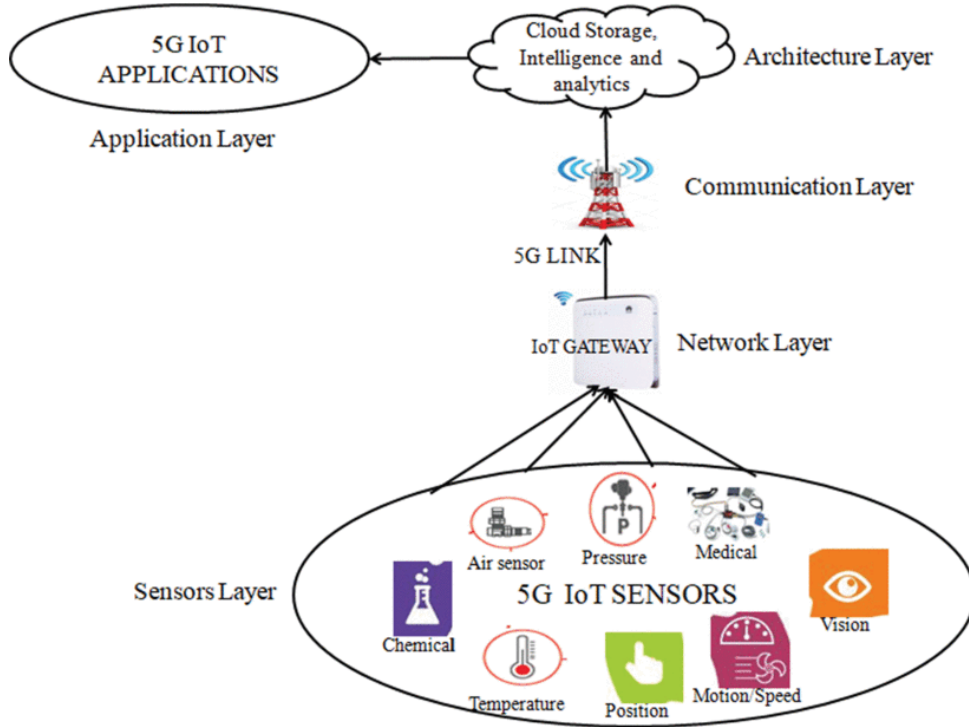


Figure 4: Five-layer IoT architecture integrated within a 5G wireless system adapted from [36].

while still benefiting from carrier-grade services [21].

The advent of 5G and beyond (B5G) networks further accelerates the potential of IIoT by supporting ultra-low latency- with target of 10ms for control plane and 0.5ms for data plane uplink and downlink [37]-, high data throughput, and secure large-scale deployments[21]. While IoT and IIoT present opportunities, they also raise significant security concerns. Challenges include device authentication, data integrity, data privacy preservation, and resistance against DoS and man-in-the-middle attacks.

2.2.2 5G system reference mode

5G mobile networks mark a significant evolution in wireless communication, engineered to support eMBB, URLLC, and mMTC [35]. To meet these diverse service requirements, 5G adopts a service-based architecture (SBA) within the 5G Core (5GC), enabling modular network functions, flexible deployment, and efficient service orchestration. Control plane network functions communicate exclusively through standardized service-based interfaces. The 5GC also provides native support for secure device registration, mutual authentication, mobility management, and traffic handling, delivering carrier-grade reliability and security essential for both consumer and industrial applications.

The non-roaming 5GC network architecture is illustrated in Fig. 5. It shows the NFs essential for non-roaming services and includes the following key components:

- **Mobility Management Function (AMF)** is responsible for UE registration, con-

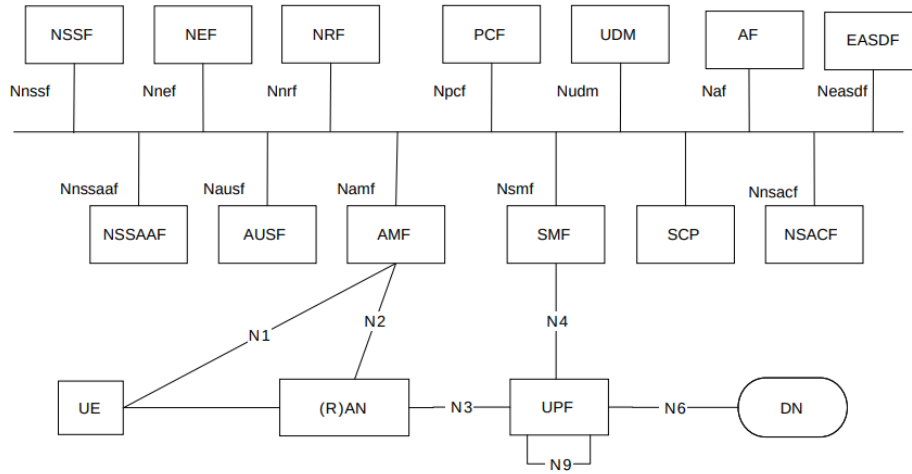


Figure 5: Non-roaming reference 5G Architecture. Adapted from [35]

nection management, access control, and mobility management. Furthermore, it integrates the Network Slice Selection Function (NSSF), which determines the appropriate network slice for a given service or subscriber.

- **Session Management Function (SMF)** handles the establishment, modification, and release of PDU sessions in accordance with network policies. It also ensures session continuity and enforces Quality of Service (QoS) requirements, thereby enabling differentiated service delivery.
- **User Plane Function (UPF)** serves as the anchor point for user-plane traffic. Its deployment can be tailored to the service type, supporting edge computing scenarios or centralized configurations. Functionally, the UPF represents the evolution of the Serving Gateway (SGW) and Packet Data Network Gateway (PGW) in 4G networks.
- **Policy Control Function (PCF)** provides a comprehensive policy framework that governs network slicing, roaming, and mobility management. It is conceptually similar to the Policy and Charging Rules Function (PCRF) in LTE, but with enhanced capabilities for dynamic and slice-aware policy enforcement.
- **Unified Data Management (UDM)** is the central repository for subscription information and user profiles. It extends the role of the Home Subscriber Server (HSS) in LTE to cover both fixed and mobile access, ensuring consistent subscriber data management across heterogeneous access technologies.
- **Network Repository Function (NRF)** introduces a novel service registration and discovery mechanism. It allows network functions to dynamically register their capabilities and discover other NFs through APIs, thereby enabling flexible and adaptive service composition.

- **Network Exposure Function (NEF)** acts as an API gateway, exposing selected network services and capabilities to external entities, such as enterprises and partner operators. This facilitates third-party integration for functions like service provisioning, monitoring, and policy control.
- **Authentication Server Function (AUSF)** provides authentication services for UEs, ensuring secure access to the network. Its specification and security procedures are defined by 3GPP SA Working Group 3 (SA WG3). The AMF and SMF interface with user-plane nodes through N1, N2, and N4, supporting subscriber attachment, session establishment, and mobility management. The N2 and N3 interfaces, in particular, depend on how the 5G radio access network (RAN) is realized, making them tightly coupled with the underlying RAN architecture.

Together, these functions form the core of the 5G service-based architecture. Additional functional entities exist to support charging, exposure, analytics, and network management but are outside the focus of this discussion.

2.3 Security Frameworks in 5G for IoT

This section briefly introduces the 5G security architecture, summarizes the two main authentication methods (5G-AKA and EAP-AKA'), and then discusses security for non-3GPP access and its integration with private 5G cores.

2.3.1 5G security architecture

Security is a cornerstone of 5G, particularly given its role in supporting mission-critical IIoT applications. Fig. 6 illustrates the overview of 3GPP 5G security architecture. The architecture is organized into six security domains, each addressing a specific aspect of end-to-end protection across the system [38]:

- **Network access security (I):** Ensures secure mutual authentication and protection of user traffic over both 3GPP and non-3GPP access networks, safeguarding the radio interface against radio interface threats, such as eavesdropping or jamming.
- **Network domain security (II):** Protects signaling and user plane communication between network entities through secure inter-node communication.
- **User domain security (III):** Secures user access to mobile equipment, preventing unauthorized device usage.
- **Application domain security (IV):** Enables secure communication between applications in user and provider domains (outside the scope of [38]).
- **SBA domain security (V):** Introduced in 5G to protect service-based architecture (SBA) communication, including network function discovery, authorization, and service interface security.

- **Visibility and configurability of security (VI):** Provides users with transparency and control over active security features.

In the 5G security architecture, primary authentication as part of network access security is performed using the 5G-AKA protocol, which provides mutual authentication between the UE and the network. In parallel, the Extensible Authentication Protocol-based AKA' (EAP-AKA') method has been introduced to provide additional flexibility, particularly in non-3GPP access scenarios. Both protocols aim to ensure confidentiality, integrity, and resistance to common security threats, while meeting the performance requirements of next-generation networks.

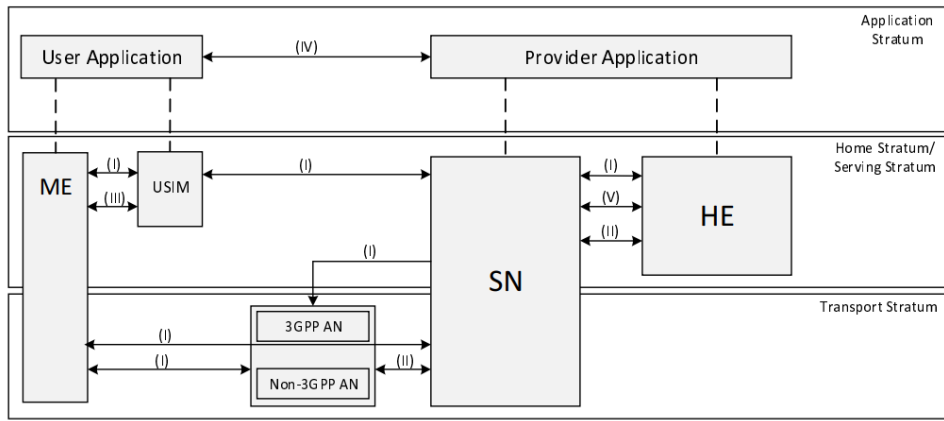


Figure 6: 5G Security Architecture Overview. Adapted from [38]

2.3.2 EAP-AKA' and 5G-AKA

The Extensible Authentication Protocol (EAP) [39] is a generic authentication framework designed to support multiple authentication mechanisms over a common message exchange structure. Rather than prescribing a specific authentication algorithm, EAP defines a standardized request/response format that can encapsulate a wide range of authentication methods, known as *EAP methods*. EAP operates between three logical entities: the *supplicant* (end device), the *authenticator* (e.g., access point or gateway), and the *authentication server* (e.g., AAA server or 5G core network). The authenticator acts as a pass-through relay, forwarding EAP messages between the supplicant and server without interpreting the method-specific content.

An EAP exchange follows a simple message pattern:

1. **EAP-Request/Identity** - The authenticator requests the identity of the supplicant.
2. **EAP-Response/Identity** - The supplicant replies with a claimed identity.
3. **Method-Specific EAP Messages** - Based on the chosen EAP method, authentication proceeds using challenge-response and key derivation steps.

EAP is widely deployed in IEEE 802.1X-based access control for wired and wireless networks, including Wi-Fi (IEEE 802.11), Ethernet edge networks, VPN systems, and cellular interworking architectures. In cellular environments, EAP is used as an alternative to native 3GPP authentication procedures and plays a central role in 5G security, especially for non-3GPP access integration such as Wi-Fi or Ethernet offloading into the 5GC.

Common EAP methods include certificate-based authentication (EAP-TLS), symmetric-key SIM-based authentication (EAP-SIM, EAP-AKA), and public-key optimization schemes (EAP-TLS-POK).

EAP-TLS-POK [40] is a lightweight public-key authentication protocol built on Transport Layer Security (TLS), specifically designed for constrained devices such as IoT/IoT hardware. Unlike classical EAP-TLS, which relies on X.509 certificates and a full public key infrastructure (PKI), EAP-TLS-POK replaces certificates with a proof-of-possession mechanism. The UE demonstrates ownership of a pre-provisioned asymmetric key pair using a Proof of Knowledge (POK) token derived from its private key. This design avoids certificate management overhead while retaining strong cryptographic assurance, making it suitable for secure onboarding of IoT devices without SIM credentials.

EAP-AKA [41] integrates the Authentication and Key Agreement (AKA) protocol used in 3G and 4G systems into the EAP framework. It enables SIM/USIM-based authentication over access networks such as Wi-Fi that may not natively support 3GPP signaling. The supplicant uses a long-term secret key K stored in its USIM and a shared subscriber profile stored in the Home Subscriber Server (HSS) or UDM entity to compute authentication vectors. EAP-AKA provides mutual authentication between UE and network, derivation of ciphering and integrity keys, and protection against replay and man-in-the-middle attacks.

However, EAP-AKA inherits some limitations from earlier 3G specifications, such as dependence on SHA-1-based key derivation functions and limited binding between authentication keys and access networks, which may impact security when roaming or interworking across domains. These limitations were addressed by its successor, EAP-AKA'.

EAP-AKA' [42] is the upgraded version of EAP-AKA and is one of the two primary authentication methods defined in the 5G system architecture, alongside 5G AKA [38]. It is the recommended method for non-3GPP access authentication in 5G, such as 5G-Wi-Fi interworking via N3IWF.

EAP-AKA' introduces several key enhancements over EAP-AKA [42]:

- **Network-Bound Key Derivation:** Session keys are derived using a 3GPP-defined key derivation function (KDF) based on HMAC-SHA-256 that binds keys to the serving network name (SN-name). This prevents key reuse across networks and improves roaming security.
- **Modern Cryptographic Primitives:** Legacy SHA-1 functions are replaced by HMAC-SHA-256, improving resistance to cryptographic collision and preimage attacks.

- **Full 5G Integration:** EAP-AKA' aligns with the 5G security architecture defined in 3GPP TS 33.501, including NAS key derivation (K_{NAS}^{5G}), anchor key generation (K_{SEAF}), and key hierarchy support for access and user-plane security.
- **Backward Compatibility:** If the serving network does not support the newer KDF, the UE falls back to EAP-AKA behavior without interrupting connectivity.

The 5G authentication process using EAP-AKA' proceeds as follows:

1. UE and network exchange EAP-Request/Identity and EAP-Response/Identity messages.
2. The network issues an EAP-Request/EAP-AKA' challenge.
3. The UE verifies network authenticity and computes the RES* response.
4. Both parties derive K_{SEAF} and subsequently K_{NAS}^{5G} .
5. Authentication concludes with EAP-Success and establishment of a secure NAS context.

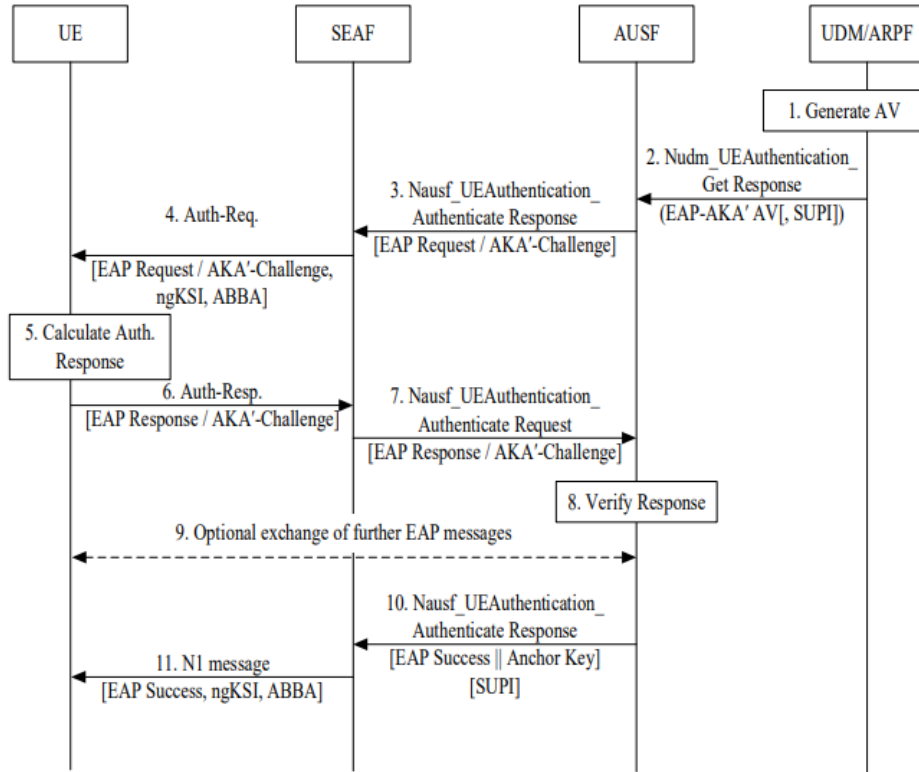


Figure 7: Authentication procedure for EAP-AKA'. Adapted from [38]

Figure 7 illustrates the authentication procedure. The process begins when the UDM or authentication Credential Repository and Processing Function (ARPF) generates an authentication vector (AV) with the AMF separation bit set, and transforms the ciphering and integrity keys (CK, IK) into CK' and IK'. This transformed AV is delivered to the AUSF, which initiates the EAP-AKA' exchange with the UE through the SEAF.

The SEAF transparently forwards the EAP-Request/AKA'-Challenge to the UE, which passes the random challenge (RAND) and authentication token (AUTN) to the USIM. The USIM verifies the freshness of the AUTN and, if successful, computes the response (RES) along with derived keys. The UE replies with an EAP-Response/AKA'-Challenge, which is validated by the AUSF. If verification fails, the authentication is aborted. Upon successful verification, both the AUSF and the UE derive the Extended Master Session Key (EMSK), from which the key K_{AUSF} is extracted. Subsequently, the serving anchor key K_{SEAF} is derived and provided to the SEAF. Using K_{SEAF} , the Subscription Permanent Identifier (SUPI), and the ABBA parameter, the SEAF computes the anchor key K_{AMF} and provides it to the AMF to establish the security context. The procedure concludes with the AUSF sending an EAP-Success message to the UE, finalizing the mutual authentication and establishing the derived keys for subsequent secure communication.

Because of its strong key binding, roaming support, and seamless interoperability with SIM-based identity management, EAP-AKA' is the de facto authentication framework for secure 5G IoT and enterprise deployments using non-3GPP access.

5G-AKA is the primary authentication and key-agreement protocol of 5G, built upon the 5G security architecture [38], is defined in the 3GPP Technical Specification Series TS 33.501 [38] and TS 33.220 [43]. 5G-AKA is an evolution of the 4G LTE AKA, enriched with subscription status checks, session context establishment, and enhanced key derivation that provides forward secrecy. The procedure of 5G-AKA is illustrated in Figure 8.

The 5G-AKA protocol ensures mutual authentication between the UE and the network while establishing cryptographic session keys. Authentication is achieved through a challenge-response exchange involving the UE, the serving network, and the home network, resulting in shared keys (K_{SEAF} , K_{AMF}) that secure NAS signaling. The formal security of 5G-AKA has been extensively analyzed and proven in 3GPP specifications [38, 44] and study [45].

The protocol provides three main guarantees: (1) *Mutual authentication* between the UE and network; (2) *Key confidentiality and integrity* for all subsequent control-plane communications; and (3) *Replay and downgrade protection* through freshness verification and the ABBA parameter.

Despite the extensive standardization effort, the intrinsic complexity of 5G-AKA—multiple nonces, timestamps, and inter-function messaging across the AMF, SMF, and UDM—renders it vulnerable to subtle design flaws that can be hard to uncover through conventional testing. Consequently, formal verification has become an indispensable tool for establishing the security properties of the protocol. In this context, [45] presents a comprehensive analysis of the 5G-AKA protocol using the ProVerif tool, a state-of-the-art symbolic verifier for cryptographic protocols.

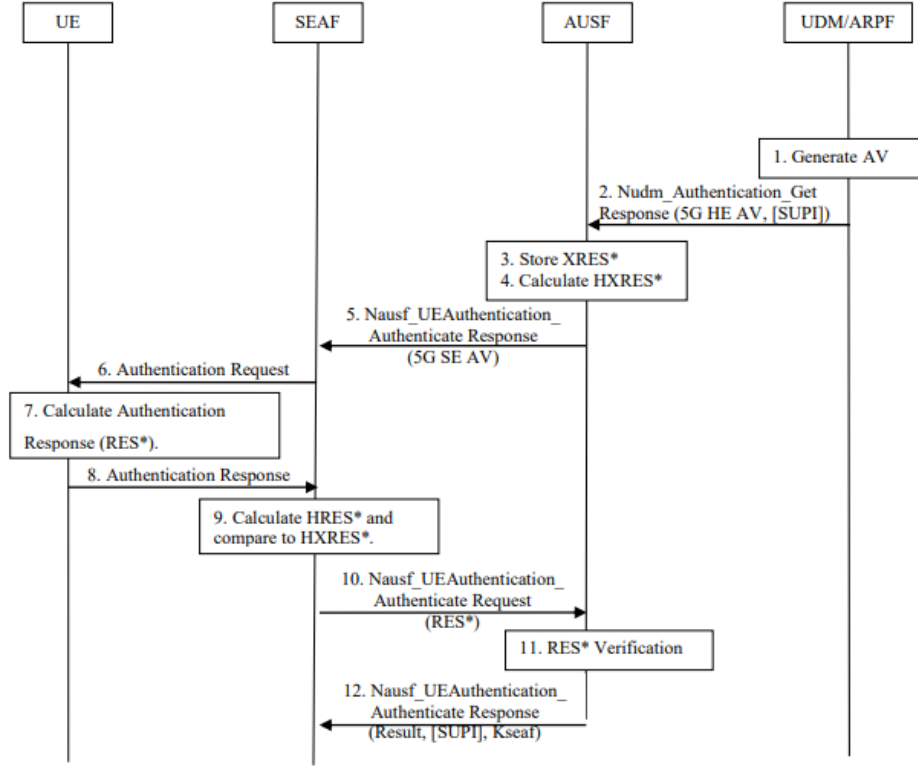


Figure 8: Authentication procedure for 5G-AKA. Adapted from [38]

The authors model the protocol in the applied π -calculus, encode the 3GPP-defined messages and key-derivation functions, and systematically verify critical properties such as mutual authentication, key freshness, and resistance to replay, man-in-the-middle, and identity leakage attacks. The verification results reported in [45] confirm that the nominal implementation of 5G-AKA satisfies the intended security goals under the Dolev-Yao adversary model.

2.3.3 Non-3GPP access security and integration with private 5G cores

The previous section described the non-roaming 5GC architecture, where UE connects to the 5G system using standardized access technology such as 5G New Radio (5G NR) through the RAN. To expand coverage, offload traffic, improve indoor connectivity, and support new services, the 5G system also supports non-3GPP access, such as WLAN networks [35]. This is especially important for IoT deployments, where many devices rely on non-cellular communication technologies. A 2024 report [46] shows that Wi-Fi accounts for 31% of all IoT connections, while cellular IoT technologies (2G, 3G, 4G, 5G, LTE-M, and NB-IoT) represent about 21% of global IoT connections.

The 5GC supports connectivity for both trusted and untrusted non-3GPP access networks. Untrusted networks connect via the Non-3GPP InterWorking Function (N3IWF), while trusted networks use the Trusted Non-3GPP Gateway Function (TNGF). These gateway functions interface with the 5GC control plane and user plane

through the standardized N2 and N3 interfaces, enabling secure session management and traffic routing. Non-3GPP access networks can advertise the public land mobile networks (PLMNs) or standalone non-public networks (SNPNs) for which they provide trusted connectivity, allowing UEs to discover and attach to networks that meet their connectivity and security requirements [35].

Figures 9 and 10 depict the service-based architecture of 5G Core networks for untrusted and trusted non-3GPP access, respectively. The primary difference between trusted and untrusted access lies in the location of the access point to which the UE connects. If the access point resides within the Home Public Land Mobile Network (HPLMN), that is, inside the 5G Core network, the access is considered trusted; otherwise, it is classified as untrusted.

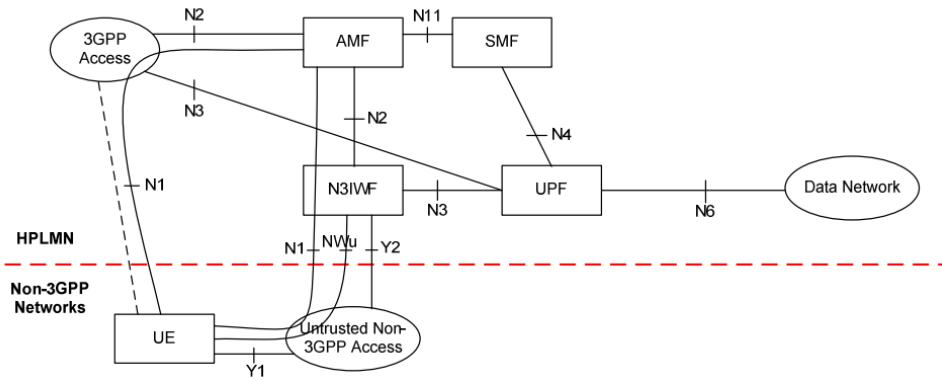


Figure 9: Non-roaming architecture for 5G Core Network with untrusted non-3GPP access. Adapted from [35]

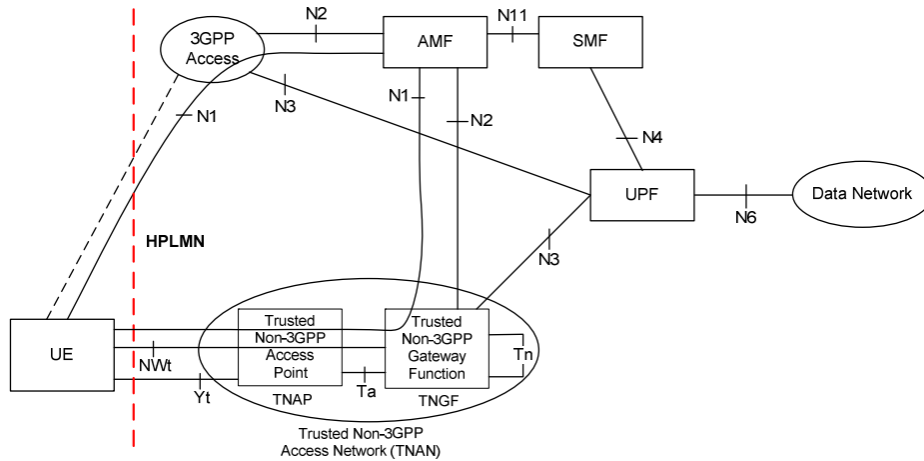


Figure 10: Non-roaming architecture for 5G Core Network with trusted non-3GPP access. Adapted from [35]

To implement the UE and gNB gateway stack for authentication and registration, it is essential to understand the network protocol stack used in the control plane. The

protocol stack is organized in layers, where each upper layer relies on and is protected by the lower ones. Higher layers handle signaling and session logic, while lower layers ensure reliable transport, framing, and physical transmission over the air interface.

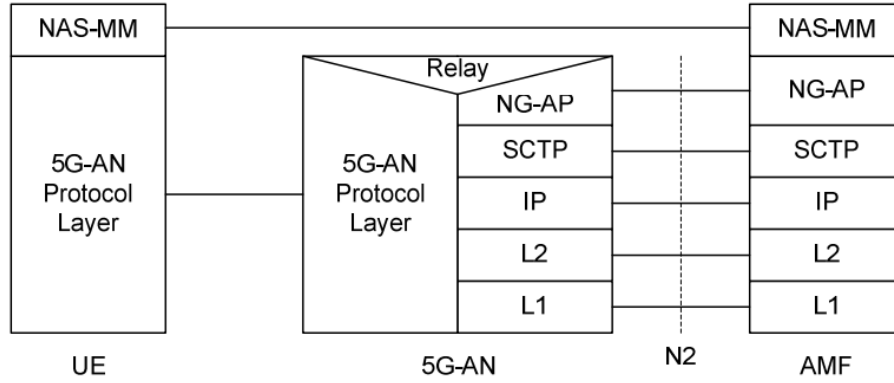


Figure 11: Control Plane protocol stack between the UE and the AMF. Adapted from [35]

Fig. 11 shows the protocol used between the UE and the AMF via the 5G access network (5G-AN). The stack includes the Non-Access Stratum Mobility Management (NAS-MM) for registration, authentication, and mobility control, the Next Generation Application Protocol (NG-AP) for signaling between the AMF and gNB, and the Stream Control Transmission Protocol (SCTP) for reliable transport of control messages. Beneath these layers, the Internet Protocol (IP) and the lower Layer 1 and Layer 2 (L1 and L2) provide end-to-end connectivity, and the access network protocol layer used between the UE and 5G-AN interface.

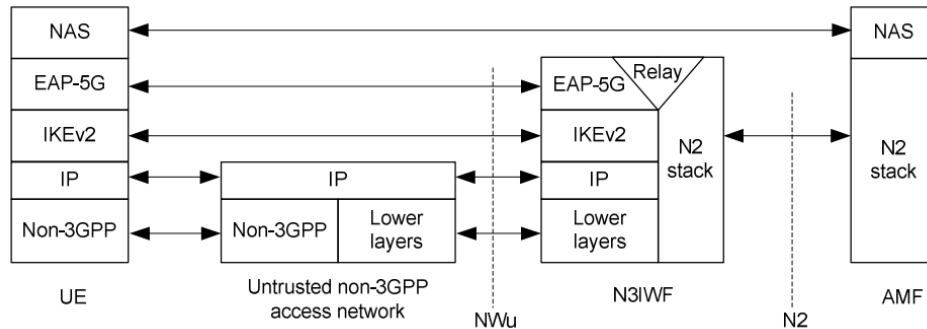


Figure 12: Control Plane protocol stack before the signalling IPsec SA is established between UE and N3IWF. Adapted from [35]

Fig. 12 illustrates the control plane protocol stack used in untrusted non-3GPP access, specifically between the UE and the N3IWF. Initial access occurs before the IPsec Security Association (SA) is established, so NAS messages are transferred using the EAP-5G or IKEv2 procedures over the underlying access network. After the IPsec SA is established, the UE creates a TCP/IP tunnel with the N3IWF to transport

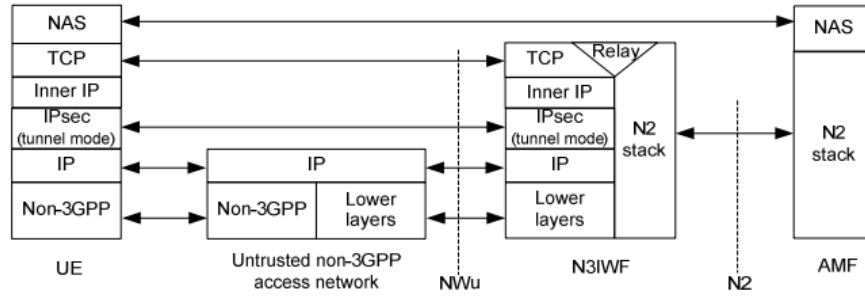


Figure 13: Control Plane after the signalling IPsec SA is established between UE and N3IWF. Adapted from [35]

NAS and PDU session management messages securely as illustrated in fig. 13. On the core network side, the AMF communicates with the N3IWF using the same SCTP and NGAP stack via the N2 interface. NAS messages between the AMF and UE are encapsulated over this secure tunnel through the N1 interface, ensuring registration, authentication, and mobility procedures are reliably delivered over untrusted access networks.

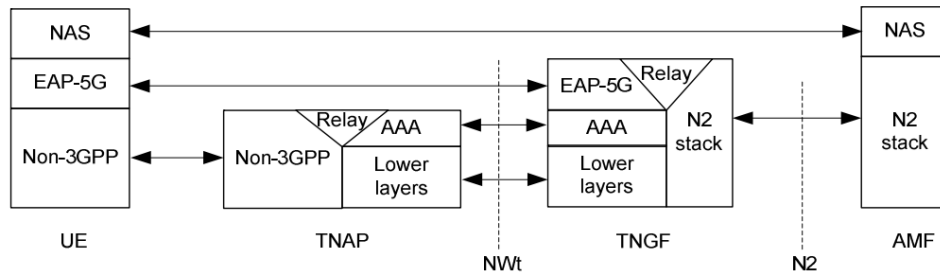


Figure 14: Control Plane protocol stack before the NWt connection is established between UE and TNGF . Adapted from [35]

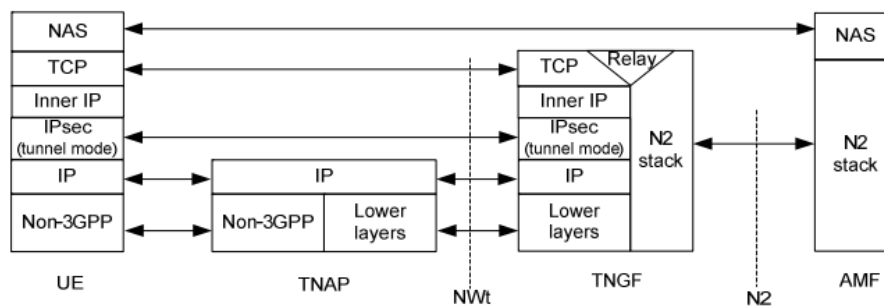


Figure 15: Control Plane after the NWt connection is established between UE and TNGF. Adapted from [35]

For trusted non-3GPP access, Fig. 14 shows the control plane protocol stack between the UE and TNGF. Before the NWt connection is established, NAS messages are encapsulated using the vendor-specific EAP-5G procedure, with the link layer supporting EAP transport and the AAA interface used between TNAP and TNGF. Once the NWt connection is active, a TCP/IP tunnel is established between the UE and TNGF, and NAS and session management messages are exchanged over IPsec as shown in fig. 15. The AMF communicates with TNGF using SCTP and NGAP via the N2 interface, while NAS messages are delivered to the UE over the N1 interface. This configuration aligns trusted non-3GPP access with the control plane procedures of untrusted access, while providing a secure channel for registration and session management.

2.4 NAS and NGAP Message Procedure in 5G-AKA

Fig. 16 illustrates the exchange of NGAP messages and the encapsulated NAS packets during the authentication and registration procedure. A detailed version of the diagram can be found in Appendix D.

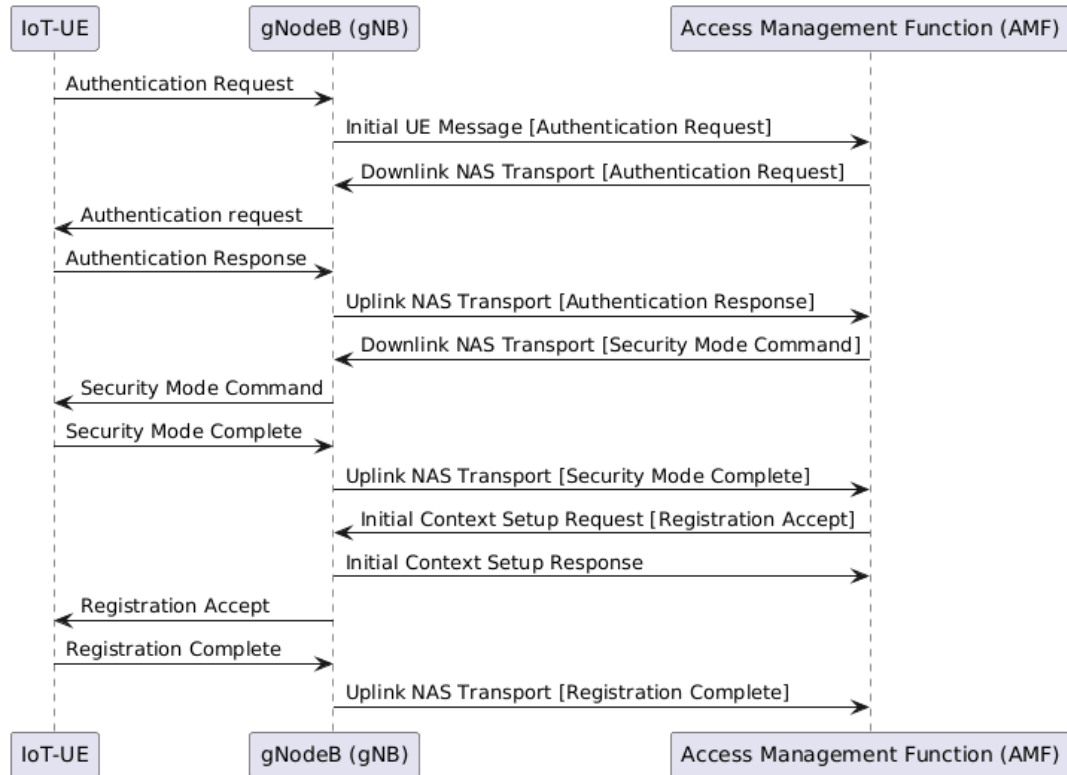


Figure 16: Sequence diagram of NGAP signaling between the gNB and AMF, showing encapsulation of NAS messages.

2.4.1 NAS Message Structure

The structure of the NAS message is standardized by 3GPP [47]. Table 2 shows an example of how a NAS packet is defined in the specification. The field IEI represents the Information Element Identifier, used to indicate optional Information Elements (IEs) present in the message. The Information Element column specifies the name or purpose of each field. The Type/Reference column provides either the type of the IE or a cross-reference to its detailed definition in the standard. The Presence column indicates whether the field is mandatory (M) or optional (O). The Format column specifies how the IE is encoded (e.g., fixed value (V), length-value (LV), type-length-value (TLV)). Finally, the Length column describes the size of the field in octets or the allowed range.

Table 2: AUTHENTICATION REQUEST message content (adapted from 3GPP TS 24.501, Table 8.2.2.1.1)

IEI	Information Element	Type/Reference	Presence	Format	Length
	Extended protocol discriminator	Extended protocol discriminator	M	V	1
	Security header type	Security header type	M	V	1/2
	Spare half octet	Spare half octet	M	V	1/2
	Authentication request message identity	Message type	M	V	1
	ngKSI	NAS key set identifier	M	V	1/2
	Spare half octet	Spare half octet	M	V	1/2
	ABBA	ABBA	M	LV	3–n
21	Authentication parameter RAND	Authentication parameter RAND (5G authentication challenge)	O	TV	17
20	Authentication parameter AUTN	Authentication parameter AUTN (5G authentication challenge)	O	TLV	18
78	EAP message	EAP message	O	TLV-E	7–1503

In general, NAS messages follow a common structure. At the start, the Extended Protocol Discriminator identifies the protocol family (e.g., 5G mobility management or session management) so that the message can be correctly interpreted. The Security Header Type specifies whether the message is plain, integrity protected, or both encrypted and integrity protected. For secured messages, the plain NAS payload is placed after the Message Authentication Code (MAC) and sequence number fields. The Type field uniquely identifies the purpose of the plain message (for example, Authentication Request or Registration Accept). Some messages also contain a Spare

Half Octet, which is reserved padding space that ensures byte alignment of subsequent fields.

This standardized structure ensures interoperability between different implementations of UEs, gNBs, and 5G Core components by defining an unambiguous format for control-plane signaling messages such as the Authentication Request.

2.4.2 NAS Message Exchange

The function of the IoT-UE stack includes constructing outbound NAS packets as well as parsing and processing received NAS messages from the core (i.e., AMF). Fig. 17 shows an exchange of 5GS mobility management messages implemented in the stack. Since NAS messages may contain different optional Information Elements (oIEs) depending on the signaling context, the IoT-UE stack ensures that all necessary oIEs are correctly included in transmitted messages, while also parsing and handling optional IEs received from the Core during registration and authentication.

Table 3 summarizes the Information Elements (IEs) involved in the procedure, outlining their purpose, function, and role.

Table 3: Summary of NAS messages during 5G Registration and Authentication

NAS Message	Key IEs	Purpose
Registration Request	ngKSI, 5GS Mobile Identity	Identify security context; UE subscription verification
Authentication Request	RAND, AUTN, ABBA	Random challenge and network authentication
Authentication Response	RES*	UE response to challenge
Security Mode Command/Complete	Selected NAS algorithms, UE security capability	Negotiate integrity/ciphering parameters
Registration Accept/Complete	5G-GUTI, Allowed NSSAI/DNN	Confirm registration and assign identifiers

A detailed breakdown of all IEs in the Registration and Authentication NAS messages is provided in Appendix B.

2.4.3 NGAP Message Structure

NGAP (Next Generation Application Protocol) is the control-plane signaling protocol exchanged over SCTP between the AMF and the gNB. In contrast to NAS, NGAP is specified using Abstract Syntax Notation One (ASN.1). ASN.1 provides a formal description of data structures, their types, and how they relate to one another. Messages defined in ASN.1 are transmitted in a binary format using Packed Encoding Rules (PER) and can be unambiguously decoded by the receiver.

For example, the element `UserLocationInformation` is defined as a CHOICE, meaning that only one of the listed alternatives can be present in a given message.

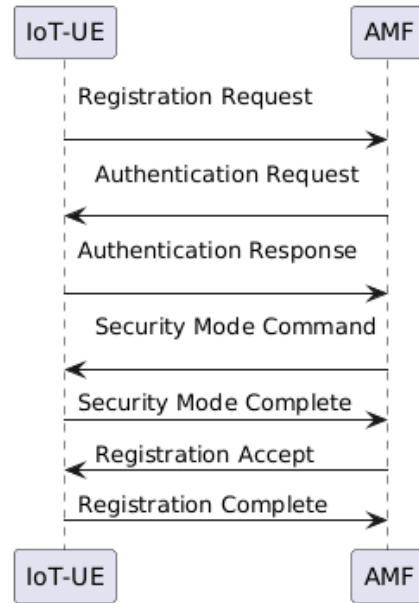


Figure 17: Sequence diagram of the authentication and registration procedure NAS flow between IoT-UE and AMF.

```

UserLocationInformation ::= CHOICE {
    userLocationInformationEUTRA      UserLocationInformationEUTRA ,
    userLocationInformationNR         UserLocationInformationNR ,
    userLocationInformationN3IWF      UserLocationInformationN3IWF ,
    choice-Extensions                 ProtocolIE-SingleContainer
}
  
```

A second example is the generic definition of an Information Element (IE), which uses the SEQUENCE keyword. A SEQUENCE groups several fields together, each with a name, type, and constraints.

```

ProtocolIE-Field ::= SEQUENCE {
    id          ProtocolIE-ID ,
    criticality Criticality ,
    value       ProtocolIE-SingleContainer
}
  
```

By using ASN.1 definitions, the structure of each NGAP message can be clearly understood and its elements accurately serialized for transmission. Figure 18 illustrates how an NGAP message is represented in tabular form within the 3GPP specification. In addition to the IE or group name, the tables include supporting fields that ensure correct message interpretation and reliable network operation. The *Presence*, *Range*, *IE type and reference*, *Semantics description*, and *Criticality* columns collectively define how each IE is processed and how errors or unknown elements are handled by the receiver.

IE/Group Name	Presence	Range	IE type and reference	Semantics description	Criticality	Assigned Criticality
Message Type	M		9.3.1.1		YES	ignore
AMF UE NGAP ID	M		9.3.3.1		YES	reject
RAN UE NGAP ID	M		9.3.3.2		YES	reject
NAS-PDU	M		9.3.3.4		YES	reject
User Location Information	M		9.3.1.16		YES	ignore
W-AGF Identity Information	O		OCTET STRING	Containing the <i>WAgfInfo</i> IE specified in TS 29.510 [36].	YES	reject
TNGF Identity Information	O		OCTET STRING	Containing the <i>TngfInfo</i> IE specified in TS 29.510 [36].	YES	reject
TWIF Identity Information	O		OCTET STRING	Containing the <i>TwifInfo</i> IE specified in TS 29.510 [36].	YES	reject

Figure 18: Example of the NGAP message UPLINK NAS TRANSPORT as standardized in 3GPP specifications.

2.5 IoT Security Enhancements Beyond 5G

Conventional IoT security mechanisms alone are insufficient to meet the unique requirements of IIoT, such as low-latency communication, lightweight cryptography, and scalability in distributed environments. To address these challenges, recent research has focused on domain-specific security mechanisms, including Intrusion Detection Systems, blockchain-enabled IIoT, as well as mechanisms to enhance the cybersecurity and resilience of 5G- and beyond-enabled IoT systems.

IDS techniques monitor network traffic and detect anomalies or malicious behaviors using statistical, signature-based, or increasingly, machine learning methods to enhance detection accuracy and adaptability in dynamic environments [4]. Recent studies have demonstrated effective IDS designs using ensemble learning and optimization-based methods to improve precision and scalability in IIoT contexts [7–9]. Meanwhile, blockchain has gained attention as a decentralized and tamper-resistant framework for ensuring trust, integrity, and secure data management in industrial systems [10]. Various implementations have applied blockchain for secure threat auditing, edge-assisted data protection, and lightweight consensus for mobile or resource-constrained devices [11–13]. Although these approaches show strong potential to complement 5G-based IoT security, they also introduce challenges such as computational overhead, latency, and limited compatibility with existing 5G core authentication and management protocols [48].

There have been a significant number of studies proposing mechanisms to enhance the cybersecurity and resilience of 5G- and beyond-enabled IoT systems. A recent systematic literature review surveyed 79 research articles in this area [14], highlighting several dominant categories of security mechanisms. Specifically, twelve studies investigated blockchain-based authentication and trust frameworks for 5G-enabled IoT. For example, Aljumah et al. proposed a Blockchain-Based AKA protocol for 5G networks [49], which improves mutual authentication while mitigating replay and man-in-the-middle attacks using distributed trust. In addition, twenty works

focused on applied cryptography and lightweight security primitives. These schemes employ digital signatures, hash-chain authentication, or post-quantum cryptographic key exchanges to secure massive IoT connectivity under strict latency and energy constraints. Finally, two studies adopted graph-theoretical approaches to model and assess the resilience of 5G-enabled IoT infrastructures, particularly for smart home dependency modeling and connectivity impact analysis.

Beyond software-based security mechanisms, hardware-assisted security has gained attention due to its low latency and resistance to tampering. Papadopoulos et al. [50] presented an FPGA-based implementation of the 5G EAP-AKA' authentication protocol, demonstrating that hardware acceleration can significantly improve authentication efficiency and secure key derivation in 5G systems.

3 System Design and Methodology

3.1 Design Objectives

While the solutions discussed in 2.5 demonstrate strong theoretical effectiveness in addressing security and resilience challenges, their real-world adoption remains limited. Many blockchain-based mechanisms demand substantial computational and storage resources, which are impractical for constrained IoT and IIoT devices. Similarly, post-quantum and advanced cryptographic schemes, though robust against sophisticated attacks, introduce considerable latency and processing overhead, hindering deployment in lightweight embedded systems. Graph-theoretical models, in turn, provide analytical insights but often lack direct applicability as operational security mechanisms.

To bridge this gap between theoretical security frameworks and deployable 5G IoT architectures, this work aims to design and implement a minimal yet functional UE–gNB system architecture that demonstrates secure, standard-compliant integration with a commercial 5G Core. The system focuses on offloading most control and security operations to the Core while minimizing device-side complexity. Because of this aim, and based on the limitations identified in the background review in Section 2.5, we specify the following design objectives for the proposed system architecture before detailing the implementation.

Objective 1: Lightweight and Modular Architecture. The system must maintain a minimal software footprint to ensure compatibility with low-power and embedded platforms (e.g., Raspberry Pi, Nordic nRF91xx). To achieve this, the UE and gNB gateway are designed as modular components implementing only the essential control-plane functions—registration, authentication, and session setup—while relying on the Core for heavy network management tasks.

Objective 2: Core-Dependent Security Offloading. Given the limited resources of IoT hardware, authentication, encryption, and session key derivation are delegated to the 5G Core's standardized security procedures (5G-AKA and NAS security). This offloading approach reduces computational overhead on devices while preserving compliance with 3GPP security specifications.

Objective 3: Standard Compliance and Interoperability. The system must

adhere strictly to 3GPP-defined interfaces and message formats for NAS and NGAP signaling. This ensures interoperability with commercial Core Network implementations (e.g., CumuCore 5G Core) and facilitates future integration into industrial deployments.

Objective 4: Resource-Aware Operation. The architecture should be deployable on both general-purpose and embedded systems, allowing comparative evaluation under constrained computational and memory conditions. This dual-platform testing validates the design’s scalability and efficiency.

Objective 5: Transparent IoT-to-5G Integration. The solution should demonstrate that existing IoT nodes can register, authenticate, and exchange data through a minimal gNB gateway directly with a 5G Core over standard interfaces, without requiring proprietary middleware or protocol translation.

Objective 6: Experimental Validation and Observability. The system must enable traceable and reproducible testing. Packet captures and message traces (e.g., NAS and NGAP logs) are used to verify successful authentication, registration, and security context establishment between the IoT-UE and the Core.

In summary, these objectives collectively guide the design of a deployable and cost-effective 5G-enabled IoT framework that demonstrates secure, standards-based connectivity for resource-limited devices in LAN-scale industrial environments. This section establishes the foundation upon which the subsequent implementation and evaluation are built.

In the prototype implementation, both the IoT-UE and the gNB gateway are hosted on Raspberry Pi platforms. The device reflects representative non-cellular, non-3GPP IoT and edge devices that are widely used in IoT prototyping. They provide generic IP connectivity toward the 5GC and offer a programmable environment for C-based development. At the same time, the software components are designed to be portable to more constrained platforms such as Nordic nRF91xx-based boards for future deployments.

3.2 System Architecture Overview

Figure 19 illustrates the system architecture. The system is composed of three main components: (i) the IoT-UE, representing a resource-constrained device (e.g., Nordic nRF91xx), (ii) the gNB gateway, implemented in C as a lightweight interface between the UE and 5GC, and (iii) the 5G Core, comprising the AMF, AUSF, and SMF services.

The gNB gateway establishes an SCTP association with the AMF to exchange NGAP signaling messages. The UE communicates with the gNB gateway through a simple socket interface, which forwards encapsulated NAS messages to the 5G Core. Control-plane security is provided through NAS integrity and ciphering once authentication succeeds. This modular separation allows IoT devices to leverage the Core’s native authentication and mobility management functions without implementing the full 5G radio and RRC stack.

The proposed architecture is designed with a focus on modularity, interoperability, and deployability within constrained environments. The system is divided into three

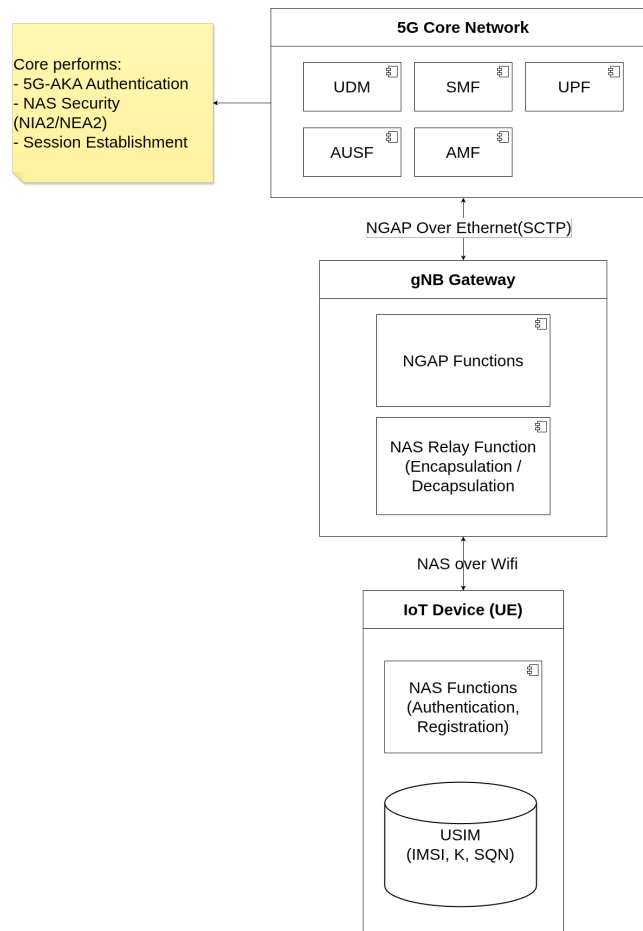


Figure 19: System Architecture

logical tiers—the IoT-UE, the gNB gateway, and the 5G Core—to balance performance, standard compliance, and implementation simplicity.

- **Modularity and Isolation:** Separating the IoT-UE from the 5G Core through a lightweight gNB gateway maintains clear boundaries between device logic and network signaling. This modular separation simplifies debugging, testing, and protocol tracing, while allowing each component to evolve independently.
- **Protocol Transparency:** The gNB gateway serves as a transparent relay for NAS-over-NGAP signaling, ensuring full compliance with 3GPP specifications without requiring proprietary interfaces or modifications to the 5G Core. This guarantees interoperability with standard Core implementations such as OAI and Open5GS.
- **Hardware Efficiency:** Deploying the UE and gateway on platforms such as the Raspberry Pi and Nordic nRF91xx enables evaluation across a range of hardware capabilities—from general-purpose edge devices to more constrained embedded systems. This separation ensures that the gNB gateway handles

signaling and processing tasks, avoiding unnecessary computational load on lightweight IoT endpoints.

- **Scalability and Maintainability:** The modular gateway structure supports easy extension to multiple UEs by maintaining separate IMSI contexts. This design avoids the complexity of monolithic architectures, which are harder to scale, maintain, and port across heterogeneous hardware.
- **Security and Reliability:** By leveraging the 5G Core’s built-in authentication and key management procedures, the system minimizes device-side security complexity while ensuring end-to-end protection. Designs that replicate authentication at the edge increase the attack surface and introduce synchronization challenges.

Overall, the selected architecture provides an optimal balance between minimalism and standard compliance, enabling secure, scalable, and practical integration of constrained IoT devices with 5G Core networks.

3.2.1 IoT User Equipment (UE)

The IoT UE represents a constrained or embedded device capable of initiating registration and authentication procedures toward the 5GC. At the logical level, the UE maintains a lightweight credential store—referred to as a software SIM (SoftSIM)—which holds essential identifiers such as the Mobile Country Code (MCC), Mobile Network Code (MNC), and Subscription Permanent Identifier (SUPI). The store also includes authentication material such as the long-term key K and operator variant configuration OP_c required for the 5G-AKA algorithm. These credentials enable the UE to participate in the standard authentication and key agreement process as defined in 2.4. In real-world deployments, hardware-based SIM or eSIM modules, such as those supported by Nordic’s nRF91xx series, would securely store these credentials and generate the Subscription Concealed Identifier (SUCI) using public key encryption. The UE thus acts as the endpoint of trust within the system, initiating 5G signaling while remaining agnostic to the transport layer beneath.

3.2.2 gNB Gateway

The gNB gateway functions as a lightweight intermediary that bridges the IoT UE with the 5G Core. It relays NAS signaling messages transparently between the UE and the AMF while preserving UE-specific identifiers such as the RAN_UE_NGAP_ID and AMF_UE_NGAP_ID. This component does not perform full RAN operations but provides sufficient logic to emulate a minimal gNB control-plane stack, facilitating authentication and registration through standard NAS/NGAP procedures.

The gateway is designed to operate within the trusted part of the 5G Core network, enabling direct AMF interaction without requiring the non-3GPP N3IWF interface, whose implementations are often incomplete or inconsistent. This simplifies deployment and ensures broad compatibility across 5GC vendors, including OpenAirInterface

(OAI) and CumuCore, and supports experimentation within private edge and industrial LAN environments.

3.2.3 5G Core Network (5GC)

The 5GC provides the control and user plane functions necessary for managing authentication, registration, and data sessions. The system supports integration with both the open-source OpenAirInterface 5GC [51, 52] and the commercial CumuCore 5GC [53]. OAI's service-based architecture (SBA) enables modular experimentation through standard APIs, while CumuCore's NFV-based design offers containerized and scalable deployments suitable for private networks. Both cores provide essential network functions such as the AMF, SMF, and AUSF for executing 5G-AKA authentication and mobility management. Additionally, CumuCore extends the architecture with advanced features including TSN and network slicing to support industrial applications [53].

Overall, the modular arrangement of UE, gNB gateway, and 5GC components allows the system to replicate real 5G authentication and registration flows, while remaining lightweight and interoperable across multiple platforms.

3.3 Control Plane Signaling and Security Mechanisms

The implemented system follows the standardized 5G control-plane signaling procedures defined in 3GPP specifications. In particular, Non-Access Stratum (NAS) messages conform to the TLV/LV encoding format specified in 3GPP TS 24.501, while NGAP messages between the gNB and the AMF are encoded using ASN.1 Packed Encoding Rules (PER). Each message exchange advances the UE through a well-defined set of protocol states, including registration initiation, authentication challenge, security mode activation, and registration completion, ensuring compliance with the 5G-AKA-based registration procedure.

To ensure robustness, the workflow incorporates message validation, state synchronization, and timeout-based retransmission handling. Security mechanisms are integrated following 3GPP TS 33.501 [38], which define the confidentiality and integrity protection of NAS signaling. Encryption is achieved using the NEA2 algorithm, based on AES in counter mode (CTR), while integrity protection is provided by the NIA2 algorithm employing AES-CMAC. These operations ensure confidentiality, authenticity, and replay protection for all control-plane exchanges between the UE and the core network.

Key derivation and authentication processes follow the Milenage algorithm set as defined in 3GPP TS 33.102 [44] and TS 33.501 [38]. The system computes the functions $f_1, f_1^*, f_2-f_5, f_5^*$ for generating the MAC, RES, and CK, IK. Higher-level derivations, including K_{AUSF}, K_{SEAF} , and K_{AMF} , are implemented to establish the secure context required for NAS message protection.

Figure 20 illustrates the end-to-end NAS security procedure and message flow between the IoT UE, gNB gateway, and AMF. Each stage highlights where key derivation, encryption (NEA2), and integrity protection (NIA2) are applied within

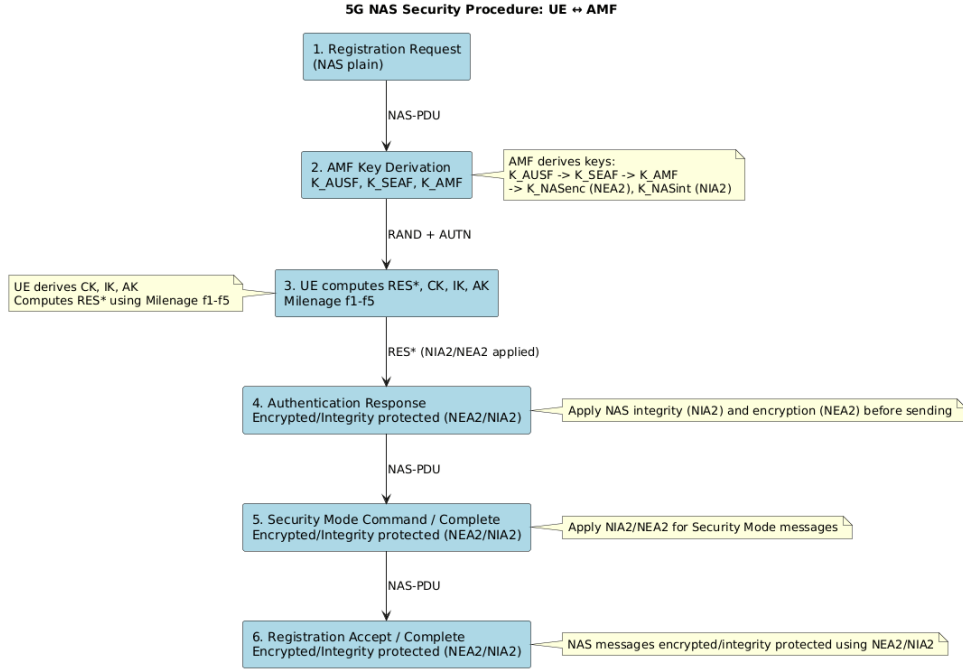


Figure 20: Overview of the 5G NAS authentication and registration procedure between the IoT UE and the AMF, showing key derivation, encryption (NEA2), and integrity protection (NIA2) steps.

the authentication and registration sequence. Together, these mechanisms ensure that the prototype accurately reproduces 5G-compliant security behavior while enabling detailed experimentation in IoT–5G integration scenarios.

4 Implementation and Evaluation

This section describes the implementation of the proposed IoT-UE and minimal gNB gateway and evaluates the system. First, the system setup are introduced. Then, the implementation details of the UE and gateway are outlined, followed by an experimental evaluation that examines functionality and interoperability with different 5G Core implementations, registration performance and basic scalability, and security related behaviour as observed from NAS and NGAP signaling traces.

4.1 System Setup

Fig. 21 presents the setup of this demonstration. The system architecture consists of IoT UEs, a minimal gNB gateway, and a commercial 5G Core. The UE runs a lightweight NAS stack in C, implementing the essential registration and session establishment procedures, with authentication based on 5G-AKA and NAS security using NIA2 and NEA2 [35, 47]. Credentials (IMSI, K, OPc, SQN) are stored locally on the device. The gNB gateway establishes NGAP signaling over SCTP, relays NAS

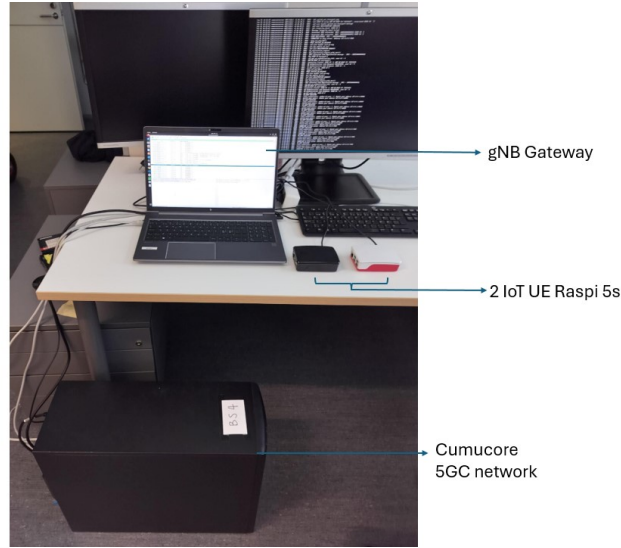


Figure 21: System setup of the demonstration, consisting of IoT UEs, a minimal gNB gateway, and a commercial 5G Core.

messages, and maintains basic UE context. The Core is provided by Cumucore Oy, that runs on a PC host and is connected with the gNB device over Ethernet. Raspberry Pi devices act as UEs, connecting via Wi-Fi through the gNB node to achieve full end-to-end registration with the 5G Core. The software implementation of the IoT UE and gNB gateway is organized into modular components to facilitate maintainability and clarity. The UE stack includes a NAS message encoder/decoder, a security module implementing 5G-AKA and NAS integrity/ciphering (NIA2/NEA2), and a state machine managing registration, authentication, and security procedures [35]. The gNB gateway acts primarily as a transparent relay, encapsulating NAS messages into NGAP PDUs and forwarding them over SCTP to the AMF while maintaining UE context for signaling purposes.

In the demo setup, the gNB gateway establishes an NGAP association with the AMF and acts as a transparent relay for NAS signaling between the IoT UE and the Core. The IoT UE generates and transmits the required NAS messages (Registration Request, Authentication Response, Security Mode Complete, and Registration Complete), while the gNB maintains the minimal UE context and forwards Core-initiated procedures such as Initial Context Setup and PDU Session Setup.

All signaling exchanges were captured and analyzed in Wireshark, as shown in Fig. 22. Wireshark automatically recognizes, classifies, and decodes standard NAS and NGAP messages, and the successful decoding of all captured packets confirms the correctness of the protocol implementation. The traces clearly show the UE authentication challenge (RAND, AUTN), the computed response (RES*), the activation of NAS security, and the establishment of a PDU session. This provides a transparent view of how a constrained IoT device interacts with the 5G Core through the minimal gNB gateway.

Fig. 23 demonstrates the network architecture of the implementation with an

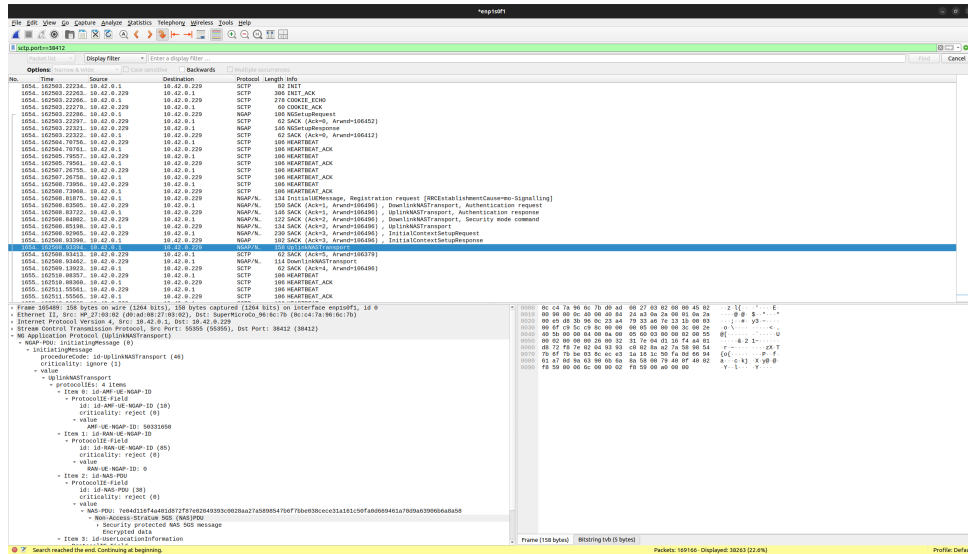


Figure 22: Wireshark trace of NAS and NGAP signaling between the IoT UE, gNB gateway, and 5G Core.

additional established data plane with Wireguard VPN tunnel between the 2 Raspberry Pis. This can be done using the encrypted NAS channel where the 5G core transports the VPN configuration to the 2 UEs. The configuration is only exposed to the UE that have the ciphering and encrypting key derive from KAMF.

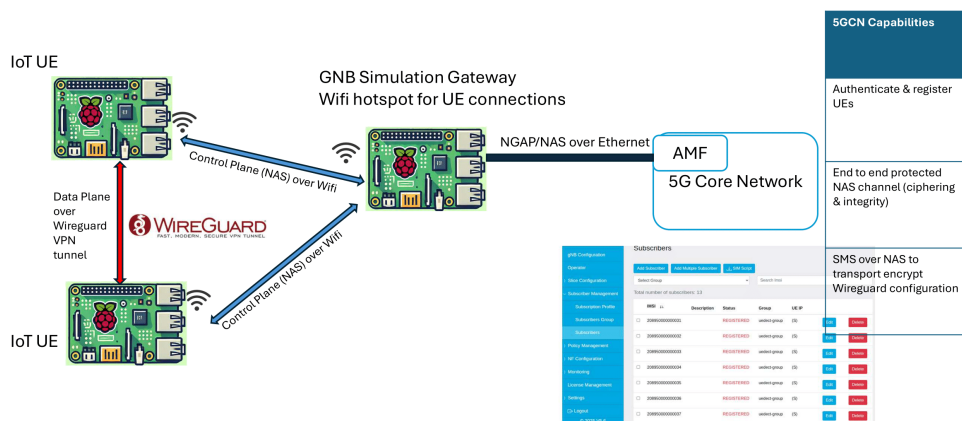


Figure 23: System architecture of the demonstration, consisting of IoT UEs stack, a minimal gNB gateway, and a commercial 5G Core.

4.2 Implementation Details

4.2.1 UE implementation and authentication handling

In the system, the IoT UE maintains a lightweight SoftSIM implemented as a local credential database. The store contains the essential identifiers and authentication material needed to execute the 5G-AKA procedure. In the code, the information is stored in a global structure called UE.

For example, the UE stores the Mobile Country Code (MCC = 208) and Mobile Network Code (MNC = 95), together forming the PLMN identifier. The Mobile Subscriber Identification Number (MSIN) is generated programmatically and combined with the PLMN to form the SUPI. The database schema also includes the long-term key K and operator variant algorithm configuration OP_c , which are required inputs for the Milenage functions. For demonstration purposes, these values are statically assigned in the implementation. Additional fields include the International Mobile Equipment Identity Software Version (IMEISV), session counts for uplink and downlink NAS security, and the Single Network Slice Selection Assistance Information (S-NSSAI) with slice/service type (SST = 222) and slice differentiator (SD = 00007b). The default Data Network Name (DNN) is set to ‘default’.

The configuration is for seamless operation with OAI-5GCN, which database is already populated with such subscribers’ configuration. CumuCore 5G core can be configured with the same configuration. Therefore, we can run the same IoT UE+gNB simulation stack with both cores. This eases up the developing and testing process.

The SUPI is formatted as a concatenation of MCC, MNC, and MSIN. In real deployments, this permanent identifier would be protected by means of SUCI generation, using the UE’s physical SIM or eSIM to apply public key encryption according to 3GPP TS 33.501. The nRF91xx chipset series, for example, supports both nano-SIM and embedded SIM (eSIM), allowing secure storage and SUCI generation in hardware.

Table 4 summarizes the key fields in the credential store.

Table 4: IoT UE credential store fields in the prototype.

Field	Example value
MCC / MNC	208 / 95
MSIN	0000001234
SUPI	208950000001234
Key K	0c0a34601d4f07677303652c0462535b
OP_c	63bfa50ee6523365ff14c1f45f88737d
IMEISV	35609204079514
S-NSSAI (SST/SD)	222 / 00007b
DNN	default

The authentication of IoT UEs follows the 5G-AKA procedure, relying on the 3GPP-specified Milenage algorithms (functions $f1$ - $f5$) and the standardized HMAC-SHA256 KDF. The complete pipeline ensures both mutual authentication and the

derivation of session keys.

Upon receiving an **Authentication Request**, the UE parses the included parameters: **RAND**, **AUTN** (detailed structure can be observed in [B](#)), and **ABBA**. The response is computed as follows:

1. Using $f2-f5$ with inputs $(K, OPc, RAND)$, the UE derives CK, IK, AK , and RES .
2. The sequence number is recovered as $SQN = (SQN \oplus AK) \oplus AK$.
3. The network is authenticated by recomputing $MAC_A = f1(K, OPc, RAND, SQN, AMF)$ and verifying it against the received MAC in **AUTN**.
4. If validation succeeds, the UE derives the anchor key material: K_{AUSF}, K_{SEAF} , and K_{AMF} using the HMAC-SHA256 KDF.
5. Finally, the UE computes the enhanced response $RES^* = KDF(RAND, RES, CK, IK)$, which is sent back to the AMF for verification.

This sequence ensures that the UE authenticates the core network and that the 5GC can in turn verify the UE, while producing fresh session keys for *NIA2* and *NEA2*. After successful mutual authentication using the Milenage functions, the UE and the 5GC derive a set of hierarchical security keys in accordance with TS 33.501. The derivation process begins with the anchor key K_{AUSF} , computed as a HMAC-SHA256-based KDF from the shared long-term key K and the random challenge $RAND$:

$$K_{AUSF} = KDF(K, FC = 0x6A, P0 = RAND, L0 = 128)$$

Subsequently, the secondary key K_{SEAF} is derived from K_{AUSF} together with the serving network name $SN - name$:

$$K_{SEAF} = KDF(K_{AUSF}, FC = 0x6B, P0 = SN-name, L0 = 128)$$

The NAS keys for encryption (K_{NASenc}) and integrity (K_{NASint}) are then obtained from K_{AMF} , which itself is derived as

$$K_{AMF} = KDF(K_{SEAF}, FC = 0x6C, P0 = SQN, L0 = 128)$$

Mathematically, the KDF is implemented via HMAC-SHA256 as follows:

1. Concatenate the function code FC with all input parameters $P_0, P_1, \dots, P_n$.
2. Apply HMAC-SHA256 with the input key (e.g., K or K_{AUSF}) to produce a 256-bit output.
3. Truncate or expand the result to the desired key length.

The HMAC-SHA256 KDF and Milenage computations are implemented in our custom C functions. The Milenage primitives ($f1$ – $f5$) produce intermediate outputs such as RES , CK , IK , and AK , which are then fed into the HMAC-SHA256 KDF to generate the hierarchical keys.

This procedure guarantees that each session has fresh, cryptographically independent keys for $NEA2$, $NIA2$, and inter-network authentication. The mathematical combination of AES-based Milenage outputs and HMAC-SHA256 KDF ensures strong one-way key derivation, resistant to replay and impersonation attacks[45].

4.2.2 gNB gateway integration and message forwarding

The gNB gateway implements the essential NG-RAN control-plane procedures required for 5G UE registration and session management, including NG-Setup, Initial Context Setup, and PDU Session Setup. During NG-Setup, the gNB establishes the NG interface with the AMF by exchanging NG-Setup Request and Response messages to configure the RAN node. Upon receiving the Initial UE Message from a connected UE, the gNB allocates the UE-specific context and resources, enabling subsequent NAS message exchanges. For data connectivity, the gNB supports PDU session creation by constructing and routing PDU Session Establishment Request and Response messages.

Each UE context maintains the current state of the UE in the registration, authentication, security, and PDU session establishment procedures. NAS messages are relayed transparently between the UE and AMF: messages from the UE are encapsulated in the appropriate NGAP container and forwarded to the AMF, while NAS messages from the AMF are decapsulated and delivered to the correct UE context. The gNB also handles connection management, including accepting new UE connections, cleaning up disconnected UEs, and retransmitting pending messages if required.

As shown in Fig. 24, the gNB operates using a minimal event-driven loop. After initializing its communication sockets, the gNB continuously polls for events from either the UE or the AMF. When receiving uplink NAS messages from a UE, it wraps them in NGAP and forwards them to the AMF, advancing the UE state as appropriate. Downlink NAS messages from the AMF are extracted and delivered to the corresponding UE. This flowchart abstracts the detailed protocol operations, highlighting the runtime control logic and NAS forwarding behavior of the gNB simulation.

4.2.3 5GC configuration and interoperability validation

The OAI 5GC is deployed on an Ubuntu laptop. Each network function is developed in a separate repository, such as `oai-cn5g-amf`, `oai-cn5g-smf`, and so on [52]. To build the complete core network, a federation repository called `oai-cn5g-fed` is provided. This repository includes the network components as submodules, a SQL database preloaded with 100 authentication subscriptions, YAML files for building network containers using Docker, and a Python script to orchestrate multiple `docker-compose` configurations corresponding to different network topologies.

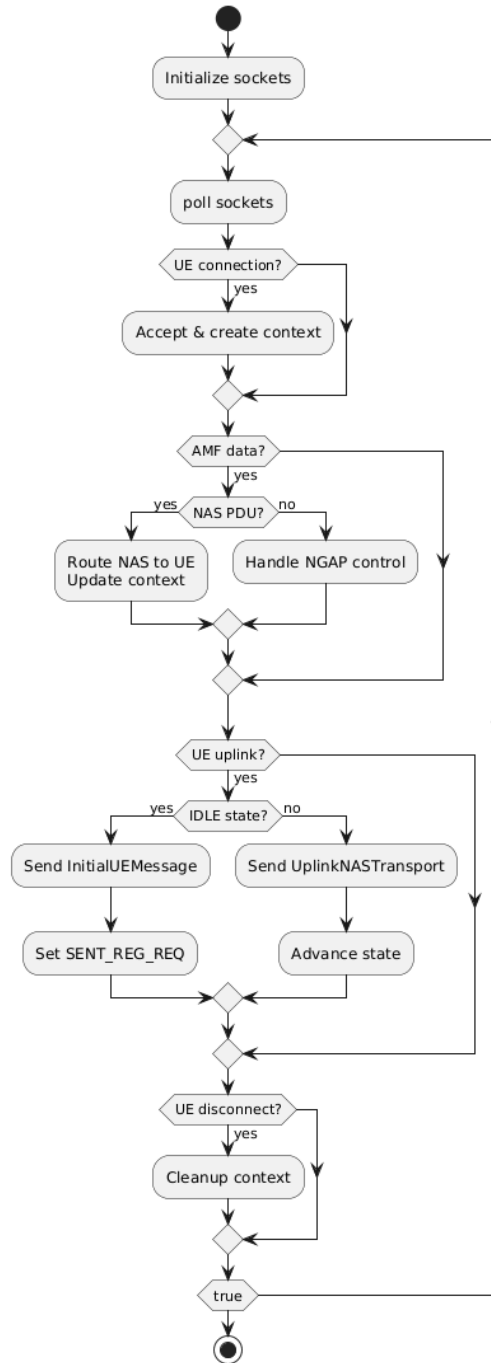


Figure 24: Simplified gNB runtime flowchart showing socket initialization, event polling, and NAS routing.

Before running `docker-compose`, the Docker images of the 5GC network components must be built. Users can either pull prebuilt images from the OAI Software Alliance Docker Hub [54] or build the images from source code. In this work, since the IoT-UE and gNB stacks mainly interact with the AMF, we opted to use prebuilt images for all components except the AMF. This allows us to modify the AMF source

code for testing, logging, or exposing internal information, significantly facilitating development. Additionally, the deployment YAML files must be reconfigured to expose the correct SCTP ports externally, as the default configuration only allows access from other Docker containers.

For this study, the provided “basic with VPP-UPF” topology was used. This topology includes the essential 5G Core Network Functions: AMF, SMF, NRF, AUSF, UDM, UDR, and the VPP-based UPF. The deployment tool launches these containerized network functions, establishes their virtual network interconnections, and optionally starts packet capture processes. During initialization, the helper script performs automated health checks to ensure that each component reaches an operational state.

After deploying the core network, the logs of the network components can be monitored to validate the authentication and registration procedures. Fig. 25 shows the logs of the `oai-amf` component after successfully registering two UEs.

```
[2025-09-15 16:49:21.194] [amf_app] [info]
```

gNBs' Information				
Index	Status	Global Id	gNB Name	PLMN
1	Connected	0x019800		208,95

UEs' Information							
Index	5GMM State	IMSI	GUTI	RAI	UE NGAP ID	AMF UE NGAP ID	Cell Id
1	5GMM-REGISTERED	208950000000031	20895010041155919653	0x00	0x03	0x04	00066c000
2	5GMM-REGISTERED	208950000000032	20895010041176007183	0x01			00066c000

Figure 25: Logs of the OAI AMF component after successful registration of two UEs.

The Cumucore 5GCN is provided by the company in the form of a binary package, accompanied by an installation bash script that automates the setup process. For installation, a dedicated PC is prepared and booted with Ubuntu Server 24.04 LTS.

After installation, unlike OAI 5GCN, which comes with a preconfigured database, the Cumucore 5GCN requires proper configuration to work with our stack. This includes setting the AMF’s network slices, tracking area codes, allowed PLMNs, selecting ciphering and integrity algorithms, configuring the IP address exposed via the Ethernet connection, adding authentication subscriptions, and activating the core with the license token provided by the company. Although some configurations require editing files directly in the core directory, Cumucore provides a user-friendly GUI accessible through a web browser, which simplifies these tasks.

Additionally, the GUI allows for the monitoring of subscribers in an organized manner, as illustrated in Fig. 26.

4.3 Evaluation and Results

This section evaluates the proposed IoT-UE and minimal gNB gateway through three categories of tests: (i) functionality, (ii) performance, and (iii) security-related behavior. The goal is to verify correct protocol operation, quantify registration latency under different conditions, and demonstrate secured data-plane communication.

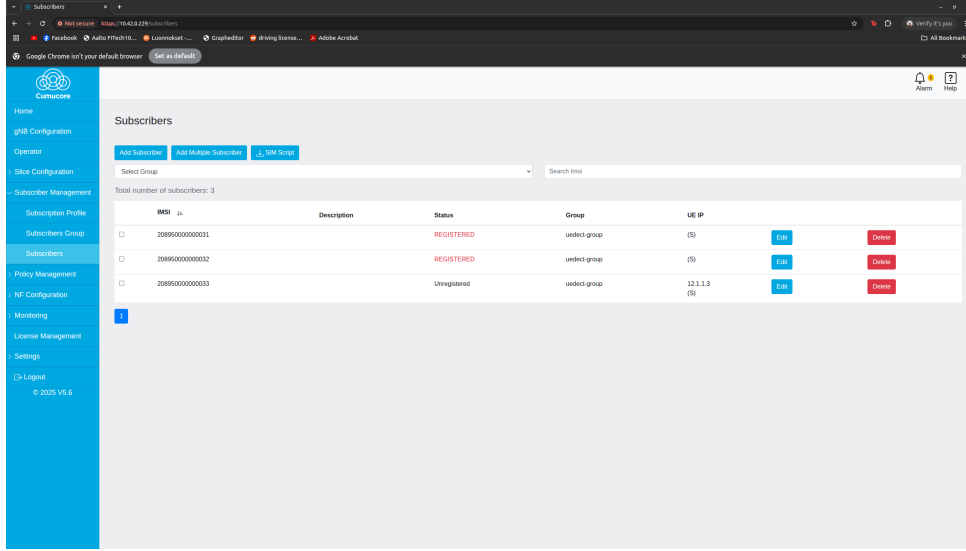


Figure 26: The GUI of CumuCore 5GCN after successful registration of two UEs.

4.3.1 Functionality Validation

Functionality tests focus on verifying that the proposed system correctly executes the 5G registration and authentication workflow. Using Wireshark, all NAS and NGAP messages exchanged between the IoT-UE, the gNB gateway, and the 5G Core were captured and automatically decoded. This confirms the correctness of message encoding, UE identifier handling, and the overall protocol state machine.

Figure 27 shows representative traces. The top portion illustrates the full 5G-AKA procedure, including the authentication challenge, the UE response, the activation of NAS security, and subsequent encrypted NAS exchanges. The lower portion shows encrypted WireGuard packets exchanged between two IoT UEs, validating data-plane functionality after successful registration.

4.3.2 Performance Evaluation

Performance tests measure end-to-end registration and authentication time under different hardware platforms, 5G Core implementations, and UE scaling conditions.

We evaluate three parameters:

- **UE scaling:** A single UE performs 100 registrations, and 100 UEs (100 IMSIs) register sequentially.
- **Hardware:** (i) Ubuntu laptop and (ii) Raspberry Pi 5.
- **5G Core:** CumuCore 5GC (commercial) and OAI 5GC (open-source).

Figure 28 presents average registration delays for all configurations. Measurement labels follow the format <5GCN>_<UE-scale>_<platform>. Table 5 provides detailed statistics (mean, standard error of the mean (SEM), standard deviation, and

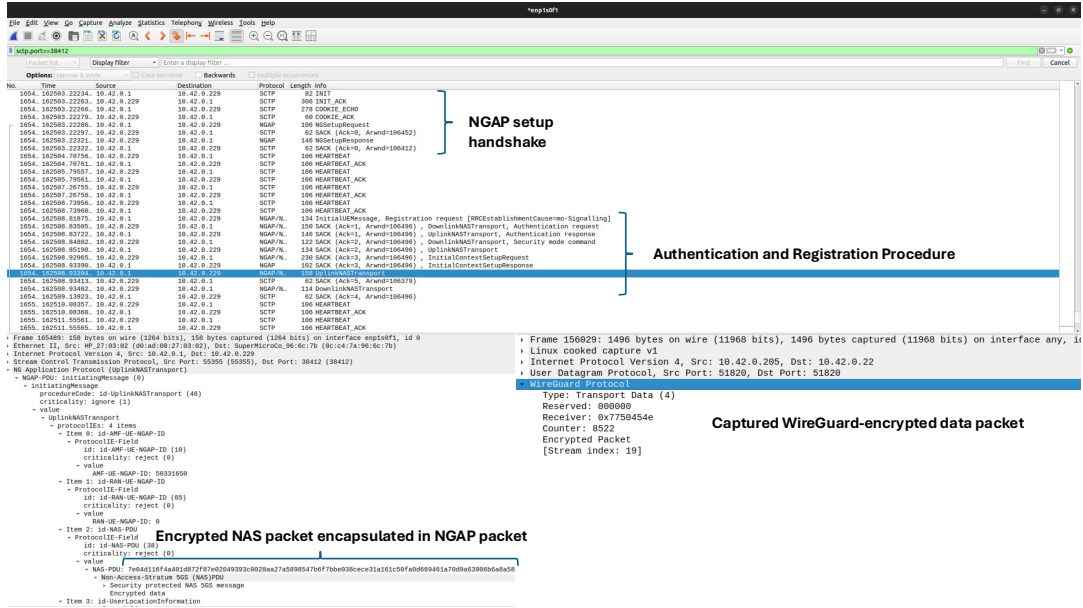


Figure 27: Wireshark captures illustrating 5G control-plane signaling (5G-AKA and encrypted NAS) and secured data-plane traffic over a WireGuard VPN tunnel.

range). Table 6 additionally reports mean/min/max values normalized to the best (lowest) value in each column.

The Cumucore 5GC shows stable performance with registration delays of 0.07–0.08 s and a consistent SEM of ≈ 0.005 s across all conditions. This reflects predictable commercial-grade control-plane behavior.

The OAI 5GC exhibits shorter delays (0.02–0.03 s) with much lower variance, largely due to its simplified internal procedures and limited feature set. Both cores demonstrate full interoperability with the proposed IoT-UE and gNB gateway.

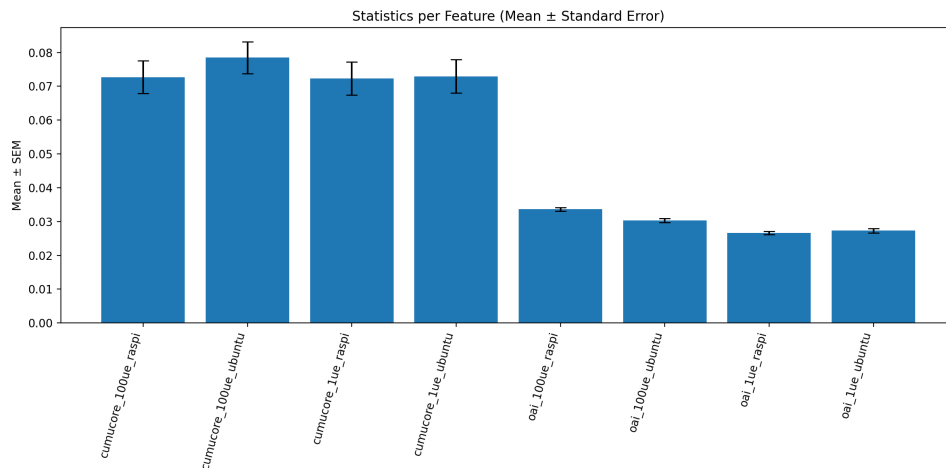


Figure 28: Mean and standard error of end-to-end authentication time across platforms and 5G Core implementations (seconds).

Table 5: Statistical summary of registration delay across configurations (milliseconds).

Metric	Mean	SEM	Std	Min	Max	n
cumucore_100ue_raspi	72.70	4.86	48.64	58.26	551.61	100
cumucore_100ue_ubuntu	78.50	4.75	47.55	55.38	535.06	100
cumucore_1ue_raspi	72.30	4.91	49.08	57.35	555.58	100
cumucore_1ue_ubuntu	73.00	4.91	49.08	50.02	550.67	100
oai_100ue_raspi	33.60	0.54	5.41	22.31	53.67	100
oai_100ue_ubuntu	30.40	0.59	5.88	21.48	59.12	100
oai_1ue_raspi	26.60	0.46	4.63	17.84	46.44	100
oai_1ue_ubuntu	27.30	0.66	6.60	19.98	62.70	100

Table 6: Mean/min/max registration delay normalized by the best (lowest) value in each column (dimensionless). Best values: mean = 26.60 ms, min = 17.84 ms, max = 46.44 ms (oai_1ue_raspi).

Metric	Mean / best	Min / best	Max / best
cumucore_100ue_raspi	2.73	3.27	11.88
cumucore_100ue_ubuntu	2.95	3.10	11.52
cumucore_1ue_raspi	2.72	3.21	11.96
cumucore_1ue_ubuntu	2.74	2.80	11.86
oai_100ue_raspi	1.26	1.25	1.16
oai_100ue_ubuntu	1.14	1.20	1.27
oai_1ue_raspi	1.00	1.00	1.00
oai_1ue_ubuntu	1.03	1.12	1.35

To test control-plane robustness, we introduce concurrent user-plane traffic using `iperf3`. As shown in Fig. 29, registration latency remains stable under load for both 5GC implementations. This demonstrates correct control–user plane separation and confirms that the gateway performs reliably in realistic deployment scenarios.

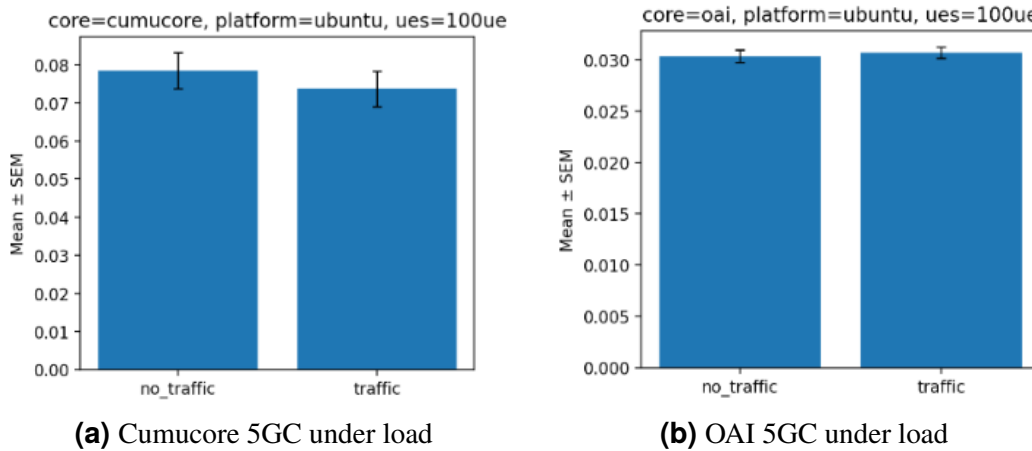
4.3.3 Security-Related Evaluation

Beyond the correct activation of NAS security, we demonstrate a secured data-plane tunnel between two IoT UEs. The gNB gateway relays WireGuard configuration messages, enabling the establishment of an encrypted VPN tunnel. All resulting traffic was verified through Wireshark captures and `iperf3` throughput tests (Fig. 30), confirming correct setup and stable encrypted communication.

4.4 Discussion

The implementation of this system involves multiple technologies and platforms, which required careful planning and integration to harmonize the overall architecture.

Figure 29: Impact of user-plane traffic on registration performance (seconds).



```

CT/dect-gnbsim$ sudo wg
interface: wg0
public key: D9sP9L1Q+pkTgokl0+b+nloJn6PxuCfMgHZ/p+14=
private key: (hidden)
listening port: 51820

peer: YBggIvD3LeilQC/35k73h6oTkM4M3wnzhuyyqQ4VYV=
endpoint: 10.42.1.205:51820
allowed ips: 10.23.0.2/32
latest handshake: 6 seconds ago
transfer: 188 B received, 272 B sent
persistent keepalive: every 25 seconds
    
```

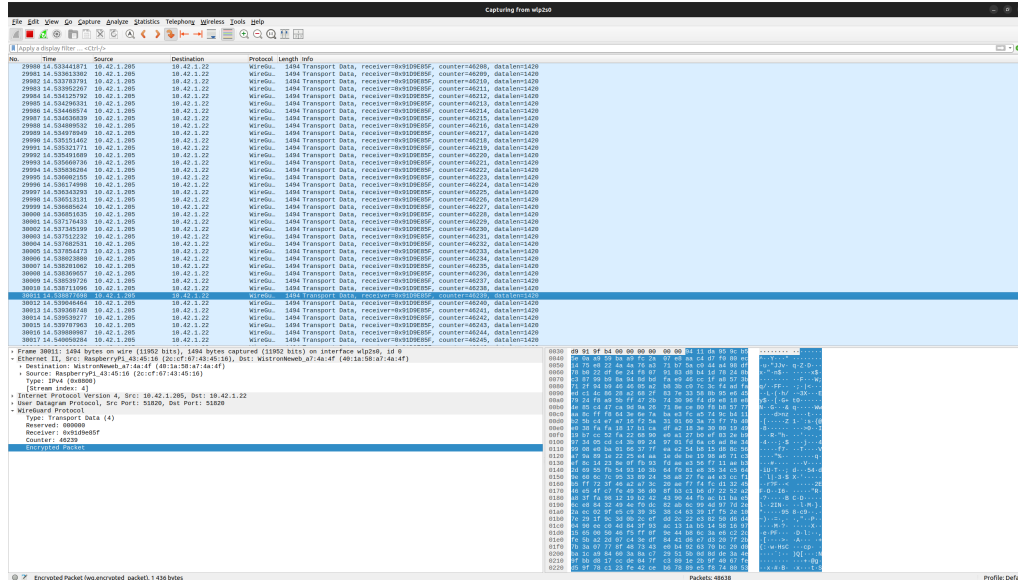
(a) WireGuard status on UE1

```

dangr@raspberrypi:~/dect-gnbsim/build$ sudo wg
interface: wg0
public key: YBggIvD3LeilQC/35k73h6oTkM4M3wnzhuyyqQ4VYV=
private key: (hidden)
listening port: 51820

peer: D9sP9L1Q+pkTgokl0+b+nloJn6PxuCfMgHZ/p+14=
endpoint: 10.42.1.22:51820
allowed ips: 10.23.0.1/32
latest handshake: 10 seconds ago
transfer: 124 B received, 180 B sent
persistent keepalive: every 25 seconds
    
```

(b) WireGuard status on UE2



(c) Encrypted WireGuard packets in Wireshark

Figure 30: Demonstration of secured data-plane tunneling with WireGuard.

The components of the implementation include various hardware devices, such as an Ubuntu laptop, a Raspberry Pi, and an Ubuntu server PC, interconnected through both wired and wireless interfaces. Ensuring seamless communication and interoperability

between these heterogeneous devices posed significant technical challenges.

These challenges, however, also provided useful insights into the practicality of the proposed design. First, the split between IoT-UE, gNB gateway, and 5GC made it easier to isolate and debug faults. Second, running the C-based NGAP stack on a Raspberry Pi shows that the control-plane design is feasible on resource-limited hardware, though CPU and I/O limits matter for larger UE counts. It has low computing and memory needs, making it accessible to low-power IoT platforms. It opens up the possibility, with no changes to the existing Core network, to use it to manage IoT nodes in IoT platforms such as LiFi, DECT-2020 NR, or backscatter. Third, successful interoperability with both commercial and open-source 5G Cores indicates that following NAS/NGAP specifications closely is sufficient to avoid vendor-specific adaptations.

Overall, the result shows that a lightweight UE and gateway architecture can deliver standards-compliant 5G integration for IoT devices without specialized 5G hardware. This suggests a general deployment pattern in which small, programmable gateways bridge heterogeneous, non-3GPP IoT platforms to private 5G networks. The same approach can be applied over different short range access technologies, such as LiFi, DECT-2020 NR, or backscatter, in industrial and campus environments. Consequently, the design provides a practical blueprint for extending 5G-grade security and control to diverse IoT deployments, while its use of inexpensive off-the-shelf hardware enables low-cost, rapid deployment.

5 Conclusion and Future Work

In this work, we demonstrated a lightweight C-based implementation of an IoT UE and gNB gateway that can register directly with a commercial 5G Core. By focusing primarily on NAS and NGAP signaling, our system avoids the complexity and overhead of full protocol stacks while maintaining full interoperability with the 5GC. The evaluation results show that even constrained devices such as Raspberry Pi and embedded development kits can reliably perform registration and authentication with carrier-grade infrastructure. This validates the feasibility of securely integrating IoT platforms into next-generation mobile networks without relying on heavyweight gateways or proprietary solutions.

Our contributions align with the long-term vision of 6G networks, particularly in enabling massive IoT connectivity, supporting heterogeneous access, and promoting security-driven orchestration. By demonstrating that lightweight devices can offload authentication, session management, and traffic control to the 5G Core, we provide a practical model for scalable and trustworthy IoT-to-5G integration in industrial and enterprise scenarios.

For future work, we plan to extend our implementation into a hybrid architecture that leverages DECT NR+ for control signaling of NAS packets between the IoT UE and gNB, while maintaining Wi-Fi for the data plane. Since DECT NR+ is designed for URLLC, this split-plane setup would ensure robust and low-latency delivery of control and management messages, leading to faster registration, authentication, and

session updates. Meanwhile, the Wi-Fi data plane would continue to support bulk traffic efficiently without congesting 5G signaling resources. Such a design would optimize spectrum usage, enhance responsiveness for time-critical IoT applications, and improve overall system robustness in industrial or enterprise deployments.

This evolution of our system not only addresses immediate deployment needs but also anticipates 6G objectives by (i) integrating diverse spectrum and access technologies, (ii) improving responsiveness and reliability for time-sensitive IoT scenarios, and (iii) ensuring scalable, secure, and resilient operation in heterogeneous environments.

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A Confidentiality Algorithm

Within the 3GPP security architecture, seven core security functions are specified, namely f_1 , f_1^* , f_2 , f_3 , f_4 , f_5 , and f_5^* . These abstract functions provide the primitives for authentication, key generation, and integrity/confidentiality protection in the Authentication and Key Agreement (AKA) family of protocols. In practice, the abstract functions f_1 – f_5^* are instantiated by concrete algorithm sets; a widely adopted instantiation is the MILENAGE algorithm set, which maps the f -functions to AES-based computations using operator-specific parameters (OP/OPc). The MILENAGE specification and its mapping of the f -functions to concrete inputs/outputs is defined in 3GPP TS 35.206 [56]. The detailed inputs and outputs of the f -functions used throughout this thesis are presented in Table A1.

Table A1: Inputs and Outputs of Functions f_1 – f_5^* adapted from [56]

Parameter	Size (bits)	Comment
Inputs to f_1 and f_1^*		
K	128	Subscriber key $K[0] \dots K[127]$.
$RAND$	128	Random challenge $RAND[0] \dots RAND[127]$.
SQN	48	Sequence number $SQN[0] \dots SQN[47]$ (for f_1^* this input is called SQN_{MS}).
AMF	16	Authentication management field $AMF[0] \dots AMF[15]$.
Inputs to f_2, f_3, f_4, f_5, and f_5^*		
K	128	Subscriber key $K[0] \dots K[127]$.
$RAND$	128	Random challenge $RAND[0] \dots RAND[127]$.
Outputs of f_1		
MAC_A	64	Network authentication code $MAC_A[0] \dots MAC_A[63]$.
Outputs of f_1^*		
MAC_S	64	Resynchronisation authentication code $MAC_S[0] \dots MAC_S[63]$.
Outputs of f_2		
RES	64	Response $RES[0] \dots RES[63]$.
Outputs of f_3		
CK	128	Confidentiality key $CK[0] \dots CK[127]$.
Outputs of f_4		
IK	128	Integrity key $IK[0] \dots IK[127]$.
Outputs of f_5		
AK	48	Anonymity key $AK[0] \dots AK[47]$.
Outputs of f_5^*		

Continued on next page

Table A1 – continued from previous page

Parameter	Size (bits)	Comment
AK	48	Resynchronisation anonymity key $AK[0] \dots AK[47]$.

For completeness, the variables used in these algorithms are defined in Table A2, while the formal symbols and operators employed in their specification are listed in Table A3. Together, these tables provide the notation and definitions required to understand the derivation and operation of the 5G AKA security functions.

Table A2: List of Variables in 5G AKA adapted from [56]

Variable	Description
AK	A 48-bit anonymity key that is the output of either of the functions $f5$ and $f5^*$.
AMF	A 16-bit authentication management field that is an input to the functions $f1$ and $f1^*$.
$c1, c2, c3, c4, c5$	128-bit constants, which are XORed onto intermediate variables.
CK	A 128-bit confidentiality key that is the output of the function $f3$.
IK	A 128-bit integrity key that is the output of the function $f4$.
IN1	A 128-bit value constructed from SQN and AMF and used in the computation of the functions $f1$ and $f1^*$.
K	A 128-bit subscriber key that is an input to the functions $f1, f1^*, f2, f3, f4, f5, f5^*$.
MAC-A	A 64-bit network authentication code that is the output of the function $f1$.
MAC-S	A 64-bit resynchronisation authentication code that is the output of the function $f1^*$.
OP	A 128-bit Operator Variant Algorithm Configuration Field used in $f1, f1^*, f2, f3, f4, f5, f5^*$.
OPC	A 128-bit value derived from OP and K and used within the computation of the functions.
OUT1–OUT5	128-bit computed values from which the outputs of $f1, f1^*, f2, f3, f4, f5, f5^*$ are obtained.
$r1, r2, r3, r4, r5$	Integers in the range 0–127 inclusive, defining the amounts by which intermediate variables are cyclically rotated.
RAND	A 128-bit random challenge input to $f1, f1^*, f2, f3, f4, f5, f5^*$.
RES	A 64-bit signed response that is the output of the function $f2$.
SQN	A 48-bit sequence number input to $f1$ and $f1^*$ (for $f1^*$, more precisely called SQN_{MS}).
TEMP	A 128-bit value used within the computation of the functions.

Table A3: List of Symbols in 5G AKA adapted from [56]

Symbol	Description
=	The assignment operator.
$\oplus$	The bitwise exclusive-OR operation.
	Concatenation of the two operands.
$E[x]_k$	Result of applying a block cipher encryption to the input value x using the key k .
$\text{rot}(x, r)$	Result of cyclically rotating the 128-bit value x by r bit positions towards the most significant bit. If $x = x[0] x[1] \dots x[127]$, and $y = \text{rot}(x, r)$, then $y = x[r] x[r+1] \dots x[127] x[0] x[1] \dots x[r-1]$.
$X[i]$	The i^{th} bit of the variable X (i.e., $X = X[0] X[1] X[2] \dots$).

The functions $f1$ – $f5^*$ rely heavily on symmetric key operations to ensure confidentiality, integrity, and authentication within the 5G AKA protocol. In particular, many of these functions make use of block cipher operations for key derivation and cryptographic transformations. The security of the 5G AKA procedure therefore depends on a robust and well-analyzed block cipher as its core primitive.

The 3GPP specifications standardize the use of the Rijndael block cipher, more commonly known as the Advanced Encryption Standard (AES), as the underlying block cipher for these functions. The following subsection briefly describes the Rijndael algorithm and its relevance to the 5G security functions.

Rijndael is a symmetric-key block cipher algorithm developed by Belgian cryptographers Joan Daemen and Vincent Rijmen. In 2001, it was selected by the U.S. National Institute of Standards and Technology (NIST) as the Advanced Encryption Standard (AES) [59]. Rijndael operates on fixed-size blocks of data and applies iterative transformations to achieve confidentiality, integrity, and authentication. It supports key lengths of 128, 192, or 256 bits; for 3GPP applications, both the block and key length are set to 128 bits.

AES accepts input as one-dimensional arrays of 8-bit bytes, which are mapped onto a 4×4 byte matrix called the *state*. The cipher key is similarly represented as a one-dimensional array of 8-bit bytes. Encryption proceeds through a series of transformations on the state, which include:

- **Byte Substitution (SubBytes):** Each byte in the state is replaced using a non-linear substitution table (S-box).
- **Row Shifting (ShiftRows):** Bytes in each row of the state matrix are cyclically shifted to the left by different offsets.
- **Column Mixing (MixColumns):** Columns of the state are transformed using matrix multiplication over a finite field $\text{GF}(2^8)$, with overflow bits reduced modulo an irreducible polynomial.
- **Add Round Key:** A round-specific subkey is XORed with the state.

The number of rounds depends on the key size:

- 128-bit key: 10 rounds
- 192-bit key: 12 rounds
- 256-bit key: 14 rounds

Rijndael derives a set of round keys from the original cipher key using a key schedule. For 128-bit and 192-bit keys, each round key is generated by cyclically rotating the previous key word, applying the S-box, and XORing with a round-dependent constant. For 256-bit keys, additional S-box transformations are applied to specific words in the expanded key. These subkeys are then used in the Add Round Key step of each encryption round.

Mathematical Representation of AES Operations Let the state at round r be denoted as $S^{(r)}$, a 4×4 byte matrix. The main AES operations can be expressed as follows:

Byte Substitution (SubBytes)

$$s_{i,j}^{(r+1)} = \text{SBox}(s_{i,j}^{(r)}), \quad 0 \leq i, j \leq 3$$

Row Shifting (ShiftRows)

$$s_{i,j}^{(r+1)} = s_{i,(j+i) \bmod 4}^{(r)}, \quad 0 \leq i, j \leq 3$$

Column Mixing (MixColumns) Each column vector $c = [s_{0,j}, s_{1,j}, s_{2,j}, s_{3,j}]^T$ is transformed using a fixed matrix over $\text{GF}(2^8)$:

$$c' = \begin{bmatrix} 02 & 03 & 01 & 01 \\ 01 & 02 & 03 & 01 \\ 01 & 01 & 02 & 03 \\ 03 & 01 & 01 & 02 \end{bmatrix} \cdot c$$

Add Round Key

$$s_{i,j}^{(r+1)} = s_{i,j}^{(r)} \oplus k_{i,j}^{(r)}, \quad 0 \leq i, j \leq 3$$

The overall transformation for one AES round can be compactly written as, and the graphical visualization can be observed in fig. A1:

$$S^{(r+1)} = \text{AddRoundKey} \circ \text{MixColumns} \circ \text{ShiftRows} \circ \text{SubBytes}(S^{(r)})$$

Finally, the complete AES encryption of a plaintext block P with key K over N rounds is:

$$C = \text{AES}_K(P) = \text{Round}_{N-1} \circ \cdots \circ \text{Round}_1 \circ \text{InitialRound}(P, K)$$

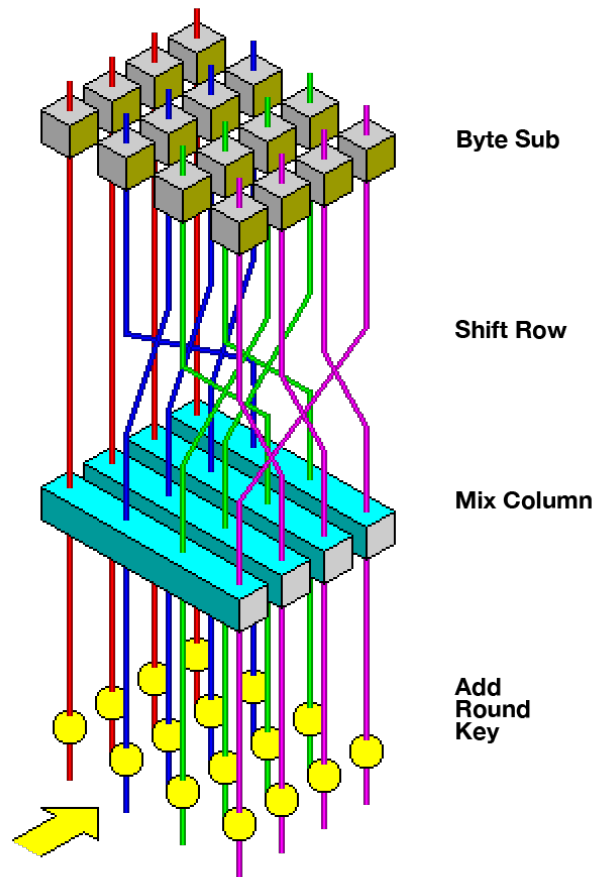


Figure A1: Visualization of the AES round function Adapted from [58]

AES provides strong security against linear and differential cryptanalysis, timing and power attacks, and opportunistic attacks. Its flexible key and block sizes, low memory requirements, and efficient implementation in both hardware and software make it suitable for a wide range of applications, including mobile devices, smart cards, and large-scale computing platforms. Compared to legacy algorithms such as DES, AES is significantly faster in software and allows for higher cryptographic security through increased key length and transformation rounds. In the context of 5G and 5G-AKA, Rijndael is used exclusively in encryption mode to derive confidentiality and integrity keys, ensuring secure exchange of authentication vectors, session keys, and subscriber data.

Keyed-Hashing for Message Authentication, commonly referred to as HMAC, is a mechanism for ensuring both data integrity and authenticity of messages [57]. HMAC combines a cryptographic hash function with a secret key, providing a secure method to verify that a message has not been altered and that it originates from a legitimate sender.

In the context of 5G-AKA, HMAC-SHA256 is used as a core component of key

derivation functions (KDFs) to generate various session and authentication keys from master secrets. HMAC operates as follows:

- A secret key is first padded to the block size of the underlying hash function (SHA-256 in this case).
- The padded key is XORed with an inner padding constant, then concatenated with the message to be authenticated.
- The hash function is applied to the result, producing an intermediate hash.
- The padded key is then XORed with an outer padding constant and concatenated with the intermediate hash.
- Finally, the hash function is applied again, yielding the HMAC output.

The resulting HMAC value is used as an input to various 5G key derivation functions to ensure that derived keys maintain integrity and resistance to tampering or replay attacks.

The detailed inputs and outputs of the KDF functions used in 5G-AKA, which rely on HMAC-SHA256, are presented in [A1](#).

Mathematically, HMAC is expressed as:

$$\text{HMAC}_K(m) = H\left((K' \oplus \text{opad}) \parallel H((K' \oplus \text{ipad}) \parallel m)\right)$$

where:

- $H(\cdot)$ is the hash function (SHA-256),
- K' is the key padded to the block size,
- $\text{ipad} = 0x36$ and $\text{opad} = 0x5C$ repeated to the block size,
- $\parallel$ denotes concatenation.

In 5G-AKA, the derived keys are computed via KDFs using HMAC-SHA256 as a pseudorandom function (PRF):

$$K_{\text{derived}} = \text{KDF}(\text{CK} \parallel \text{IK}, \text{Context})$$

Here, the concatenated CK and IK form the secret input, while Context encapsulates identifiers such as SUPI, serving network name, and algorithm identifiers. This ensures that all session and authentication keys inherit strong integrity and confidentiality properties from the master secrets.

Together, AES and HMAC form the cryptographic backbone of 5G-AKA, providing confidentiality, integrity, and authentication for secure communication between the UE and the 5G Core network.

B NAS Content

Table B1: Registration Request (UE → AMF)

Information Element (IE)	Purpose / Function
NAS key set identifier (ngKSI)	Identifies the active NAS security context between the UE and the 5G Core. In 5G Standalone (SA), it always refers to a native 5G context (KSI_AMF), ensuring the correct authentication, integrity, and encryption keys are applied.
5GS Mobile Identity	Concealed permanent identifier used for privacy protection (SUPI concealment). The core network uses this field to verify in the subscription database whether the UE is a valid subscriber.
UE Security Capability	Lists supported NAS integrity/ciphering algorithms (e.g., NIA2, NEA2).
5GS update type / DNN	Indicates requested update type or connectivity, e.g., for CIoT behaviour or SMS over NAS.

Table B2: Authentication Request (AMF → UE)

Information Element (IE)	Purpose / Function
ABBA	5G parameter used to avoid bidding-down attacks between authentication methods (e.g., ensuring consistency between 5G-AKA and EAP-AKA'). It protects against downgrade attempts by malicious intermediaries.
RAND	Random parameter generated by the core network.
AUTN	Authentication token: contains the concealed sequence number ($SQN \oplus AK$), the AMF field, and a MAC.

Table B3: Authentication Response (UE → AMF)

Information Element (IE)	Purpose / Function
RES*	UE-computed response (RES or RES*)

Table B4: Security Mode Command (AMF → UE)

Information Element (IE)	Purpose / Function
NAS selected security algorithms	Indicate the selected integrity algorithm (NIA) and ciphering algorithm (NEA).
UE Security Capability	Replay of the UE security capability IE that is sent in the Registration Request message.
IMEISV request	Core requests a 16-digit number that identifies a mobile device and its specific software version.
Additional 5G Security Information	Core may request retransmission of Registration Request.

Table B5: Security Mode Complete (UE → AMF)

Information Element (IE)	Purpose / Function
IMEISV	Send the requested parameter to the core.
NAS Message Container	The requested retransmission of the Registration Request message is included in this IE

Table B6: Registration Accept (AMF → UE)

Information Element (IE)	Purpose / Function
5GS registration result	Result of the registration process
Assigned 5G-GUTI / TAI	Temporary identifier and registration area information assigned by AMF.
Allowed NSSAI / DNN info	Slice and data network assignments approved by network.

Table B7: Registration Complete (UE → AMF)

Information Element (IE)	Purpose / Function
Optional NAS Container / IEs	Confirms completion; may carry optional configuration or status information.

C Code Snippets

Listing 1: UE authentication structure definition

```
#ifndef UE_H
#define UE_H

#include <stdint.h>
#include <stdbool.h>
#include <string.h>
#include <stdlib.h>

typedef struct
{
    char *opc;
    char *k;
    uint8_t rand[16];
    uint8_t autn[16];
    uint8_t sqn_xor_ak[6];
    uint8_t amf[2];
    uint8_t mac[8];
    uint8_t sqn[6];
    uint8_t ck[16];
    uint8_t ik[16];
    uint8_t ak[6];
    uint8_t resstar[16];
    uint8_t abba[2];
    uint8_t kint[16];
    uint8_t kenc[16];
} authparam;

typedef struct
{
    bool imeisv;
    bool rinmr;
} recv_flag;

typedef struct
{
    uint8_t pduSessionId;
    uint8_t procedureTransactionId;
} sm;

typedef struct
{
```

```

        char *sd;
        int sst;
    } snssai;

typedef struct
{
    char *imeisv;
    char *supi;
    char *msin;
    int routing_indicator;
    char *protectionscheme;
    char *dnn;
    int mcc;
    int mnc;
    uint32_t dl_count;
    uint32_t ul_count;
    authparam auth;
    recv_flag recv_flag;
    sm sm;
    snssai snssai;
} UE;

extern UE *ue;

UE *ue_init();
void ue_free(UE *u);

#endif

```

D Sequence Diagram

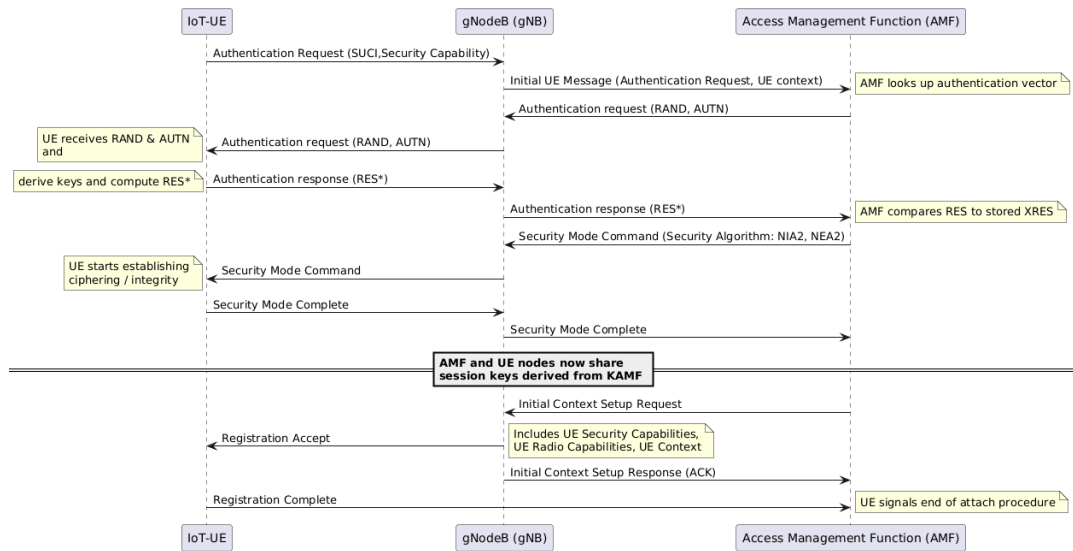


Figure D1: Sequence diagram of the exchanging data packets and executed task in details